Anonymous Technical Report

Supplementary Theoretical Results and Proofs

A Proof of Lemma 1

The main idea of the proof is inspired from the proof of [19] Theorem 3. Our main results depend on the structure of a weighted undirected graph $\mathcal{G}_t$ defined as follows.

Definition 1 (Comparison Graph $\mathcal{G}_t$). *For any $t \in [T]$, let each objective $j \in [D]$ corresponds to a vertex. For any pair of vertices $j, j' \in [D]$, there is a weighted edge between them with the weight equals to $\sum_{i\in[t]} \frac{m_i}{D^2}$. Let $L_{\mathcal{G}_t}$ denotes the Laplacian matrix of this weighted complete graph.*

Observe that for any t, $L_{\mathcal{G}_t}$ is positive semi-definite and the smallest eigenvalue of $L_{\mathcal{G}_t}$ is zero with the corresponding eigenvector given by the normalized all-one vector. Let $0 = \xi_1(\mathcal{G}_t) \le \xi_2(\mathcal{G}_t) \le \ldots \le \xi_D(\mathcal{G}_t)$ denote the eigenvalues of $L_{\mathcal{G}_t}$ in ascending order.

Proof of Lemma 1. By the loss function for MLE optimization, for any number of QE sample t, we have

$$\begin{aligned}\mathcal{L}_{\mathcal{S}_t}(\boldsymbol{c}) &= -\sum_{i=1}^{t}\sum_{j=1}^{m_i}\log\big(\frac{\exp(\boldsymbol{c}_{S_i(j)})}{\sum_{j'=j}^{m_i}\exp(\boldsymbol{c}_{S_i(j')}) + \sum_{j'\in[D]\setminus S_i}\exp(\boldsymbol{c}_{j'})}\big)\\ &= -\sum_{i=1}^{t}\sum_{j=1}^{m_i}\log\big(\frac{\exp(\boldsymbol{c}_{\sigma_i(j)})}{\sum_{j'=j}^{D}\exp(\boldsymbol{c}_{\sigma_i(j')})}\big)\end{aligned}$$

Let $\nabla\mathcal{L}_t(c)$ as the gradient of $\mathcal{L}_{\mathcal{S}_t}(\boldsymbol{c})$ w.r.t $\boldsymbol{c}$, $H_t(c)$ as the Hessian matrix of $\mathcal{L}_{\mathcal{S}_t}(\boldsymbol{c})$ w.r.t $\boldsymbol{c}$. Define $\Delta_t = \hat{c}_t^{(\mathrm{QE})} - \bar{c}^0$. It follows from the definition that Δ_t is orthogonal to the vector $\mathbf{1}$. By the definition of MLE estimator, we have $\mathcal{L}_{\mathcal{S}_t}(\hat{c}_t^{(\mathrm{QE})}) \le \mathcal{L}_{\mathcal{S}_t}(\bar{c}^0)$, and thus

$$\mathcal{L}_{\mathcal{S}_t}(\hat{c}_t^{(\mathrm{QE})}) - \mathcal{L}_{\mathcal{S}_t}(\bar{c}^0) - \nabla\mathcal{L}_t(\bar{c}^0)^\top\Delta_t \le -\nabla\mathcal{L}_t(\bar{c}^0)^\top\Delta_t \le \|\nabla\mathcal{L}_t(\bar{c}^0)\|_2\|\Delta_t\|_2,$$

where the last inequality holds due to the Cauchy-Schwartz inequality. By the Taylor expansion, there exists a $c = a\bar{c}^0 + (1-a)\hat{c}^{(\mathrm{QE})}$ for some $a \in [0,1]$ such that

$$\mathcal{L}_{\mathcal{S}_t}(\hat{c}_t^{(\mathrm{QE})}) - \mathcal{L}_{\mathcal{S}_t}(\bar{c}^0) - \nabla\mathcal{L}_t(\bar{c}^0)^\top\Delta_t = \frac{1}{2}\Delta^\top H_t(c)\Delta \ge \frac{1}{2}\xi_2(H_t(c))\|\Delta_t\|_2^2,$$

where the last inequality holds because the Hessian matrix $H_t(c)$ is positive semi-definite and $H_t(c)^\top\mathbf{1} = 0, \Delta_t^\top\mathbf{1} = 0$. Combining above results yields

$$\|\Delta_t\|_2 \le \frac{2\|\nabla\mathcal{L}_t(\bar{c}^0)\|_2}{\xi_2(H_t(c))}. \tag{7}$$

We next present two key auxiliary results used in the proof.

Lemma 6. *For any sufficiently large number of samples $t \ge 8(2e^{2B}+1)\sqrt{2DT\log\frac{T}{\rho}}$, with probability at least $1-\frac{D\rho^2}{t^2}$, we have*

$$\xi_2(H_t(c)) \ge \frac{t\overline{m}_t}{D(4e^{4B}+2e^{2B})}.$$

Lemma 7. *With probability at least $1-\frac{e^2\pi^2\rho^2}{3}$, for any $t \in [T]$, we have*

$$\|\nabla\mathcal{L}_t(\bar{c}^0)\|_2 \le 4\sqrt{\overline{m}_t t\log\frac{t}{\rho}}.$$

Combining result in Eq. 7 with above lemmas, we have with probability at least $1-\frac{(D+2e^2)\pi^2\rho^2}{6}$, for all $t \ge 8(2e^{2B}+1)\sqrt{2DT\log\frac{T}{\rho}}$, we have

$$\|\Delta_t\|_2 \le D(32e^{4B}+16e^{2B})\sqrt{\frac{\log(t/\rho)}{\overline{m}_t t}}.$$

□

A.1 Proof of Lemma 6

Proof. Define $S^{(i,j)}$ as the set of objectives contending for the j-th position in the ranking of i-th ranking sample after higher ranking objectives have been selected. Here $j \in [1,m]_i$. Let $\mathbb{1}^{(i,j)}$ denote the indicator vector for the set $S^{(i,j)}$, and let $p^{(i,j)}$ denote the the corresponding probability column vector for the selection, then the Hessian can be written as:

$$H_t(c) = \sum_{i=1}^{t}\sum_{j=1}^{m_i} H^{i,j},$$

where $H^{i,j} = \frac{1}{2}\sum_{k,k'\in S^{(i,j)}}(e_k - e_{k'})(e_k - e_{k'})^\top p_k^{(i,j)} p_{k'}^{(i,j)} = \mathrm{diag}(p^{(i,j)}) - p_k^{(i,j)} p_k^{(i,j)\top}$, with $p_k^{(i,j)} = \frac{\mathbb{1}_k^{(i,j)}\exp(c_k)}{\sum_{k'\in S^{(i,j)}}\exp(c_{k'})}$, e_x is the x-th standard basis vector.

By Claim 1 in [19], we have

$$e^{2B}H^{i,j} \succeq \frac{1}{2(D-j+1)^2}\sum_{k,k'\in S^{i,j}}(e_k - e_{k'})(e_k - e_{k'})^\top.$$

Summing over i and j, and note that $D - j + 1 \le D$, we have

$$e^{2B}H_t(c) \succeq \frac{1}{2}\sum_{k,k'\in S^{i,j}}(e_k - e_{k'})(e_k - e_{k'})^\top \frac{1}{D^2}\sum_{j=1}^{m_i}\mathbb{1}_{\{\sigma_i^{-1}(k),\sigma_i^{-1}(k')\ge j\}} := \tilde{L}_t. \tag{8}$$

Additionally, Observe that

$$\begin{aligned}\sum_{j=1}^{m_i}\mathbb{P}_c[\sigma_i^{-1}(k),\sigma_i^{-1}(k') \ge j] &= 1 + \sum_{k''\in[D]}\mathbb{1}_{\{k''\ne k,k'\}}\frac{\exp(c_{i''})}{\exp(c_i)+\exp(c_{i'})+\exp(c_{i''})}\\ &\ge 1 + \frac{m_i - 1}{2e^{2B}+1} \ge \frac{m_i}{2e^{2B}+1}.\end{aligned}$$

Recall that $L_{\mathcal{G}_t}$ is the Laplacian of $\mathcal{G}_t$ and $L_{\mathcal{G}_t} = \sum_{i=1}^t L_i$, we have

$$\begin{aligned}\mathbb{E}[\tilde{L}_t] &= \frac{1}{2}\sum_{i=1}^t\sum_{k,k'\in[D]}(e_k - e_{k'})(e_k - e_{k'})^\top\frac{1}{D^2}\sum_{j=1}^{m_i}\mathbb{P}_c[\sigma_i^{-1}(k),\sigma_i^{-1}(k')\ge j]\\ &\ge \frac{1}{2}\sum_{i=1}^t\sum_{k,k'\in[D]}(e_k - e_{k'})(e_k - e_{k'})^\top\frac{m_i}{D^2(2e^{2B}+1)} = \frac{L_{\mathcal{G}_t}}{2e^{2B}+1},\end{aligned} \tag{9}$$

Define $a_{kk'} = \frac{1}{D^2}\sum_{j=1}^{m_i}\left(\mathbb{1}_{\{\sigma_i^{-1}(k),\sigma_i^{-1}(k')\ge j\}} - \mathbb{P}_c\left[\sigma_i^{-1}(k),\sigma_i^{-1}(k')\ge j\right]\right)$, then

$$\tilde{L} - \mathbb{E}_c[\tilde{L}] = \frac{1}{2}\sum_{i=1}^t\left(\sum_{k,k'\in[D]}a_{kk'}(e_k - e_{k'})(e_k - e_{k'})^\top\right) := \sum_{i=1}^t Y_i.$$

Observe that $|a_{kk'}| \le \frac{m_i}{D^2}$ and therefore $-L_i \le Y_i \le L_i$. Furthermore, note the spectral norm $\|L_i\| = \frac{m_i}{D}$, and thus $\|Y_j\| \le \frac{m_i}{D}$. Moreover,

$$Y_j^2 = \sum_{k,k',k''\in[D]}a_{kk'}a_{kk''}(e_k - e_{k'})(e_k - e_{k''})^\top.$$

It follows that for any vector $x \in \mathbb{R}^D$,

$$\begin{aligned}x^\top Y_j^2 x &= \sum_{k,k',k''\in[D]}a_{kk'}a_{kk''}(x_k - x_{k'})(x_k - x_{k''})\\ &\le \frac{m_i^2}{D^4}\sum_{k,k',k''\in[D]}|x_k - x_{k'}||x_k - x_{k''}|\\ &= \frac{m_i^2}{D^4}\sum_{k\in[D]}\left(\sum_{k'\in[D]}|x_k - x_{k'}|\right)^2\\ &\le \frac{m_i^2}{D^4}\sum_{k\in[D]}\left(D\sum_{k'\in[D]}(x_k - x_{k'})^2\right) \qquad \text{(Cauchy–Schwarz inequality)}\\ &= \frac{m_i^2}{D^3}\sum_{k,k'\in[D]}(x_k - x_{k'})^2 = \frac{2m_i}{D}x^\top L_i x.\end{aligned}$$

Therefore, we have $Y_i^2 \le \frac{2m_i}{D}L_i \le 2L_i$, which implies $\|\sum_{i=1}^t\mathbb{E}_c[Y_i^2]\| \le \frac{2m_i}{D}\xi_D(\mathcal{G}_t) \le 2\xi_D(\mathcal{G}_t)$.

By the matrix Bernstein inequality [29], we have with probability at least $1 - \frac{D\rho^2}{t^2}$,

$$\|\tilde{L}_t - \mathbb{E}_c[\tilde{L}_t]\| \le 2\sqrt{2\xi_D(\mathcal{G}_t)\log\frac{t}{\rho}} + \frac{4}{3}\log\frac{t}{\rho}.$$

By the assumption that $\xi_D(\mathcal{G}_t) \geq C \log \frac{T}{\rho}$ for some sufficiently large C, we have

$$\|\tilde{L}_t - \mathbb{E}_c[\tilde{L}_t]\| \leq 4\sqrt{2\xi_D(\mathcal{G}_t)\log\frac{t}{\rho}}.$$

Combining Eq. 8 Eq. 9 with above result yields

$$\xi_2(H_t(c)) \geq \frac{1}{e^{2B}}\xi_2(\tilde{L}_t) \geq \frac{1}{e^{2B}}\Big(\frac{1}{2e^{2B}+1}\xi_2(\mathcal{G}_t) - 4\sqrt{2\xi_D(\mathcal{G}_t)\log\frac{t}{\rho}}\Big).$$

Recall $L_{\mathcal{G}_t} = \sum_{i=1}^t L_i$, and $\overline{m}_t = \frac{1}{t}\sum_{i=1}^t m_i$, we have the complete comparison graph $\mathcal{G}_t$ with edge weight of $\frac{\overline{m}_t t}{D^2}$, which implies

$$\xi_1(\mathcal{G}_t) = 1, \qquad \xi_2(\mathcal{G}_t) = \ldots = \xi_D(\mathcal{G}_t) = \frac{\overline{m}_t t}{D}.$$

Putting everything together, we have for any $t \geq 8(2e^{2B}+1)\sqrt{2DT\log\frac{T}{\rho}}$, with probability at least $1 - \frac{D\rho^2}{t^2}$,

$$\begin{aligned}
\xi_2(H_t(c)) &\geq \frac{1}{e^{2B}}\Big(\frac{1}{2e^{2B}+1}\xi_2(\mathcal{G}_t) - 4\sqrt{2\xi_D(\mathcal{G}_t)\log\frac{t}{\rho}}\Big)\\
&= \frac{1}{e^{2B}}\Big(\frac{1}{2e^{2B}+1}\frac{\overline{m}_t t}{D} - 4\sqrt{2\xi_D(\mathcal{G}_t)\log\frac{t}{\rho}}\Big)\\
&\geq \frac{\overline{m}_t t}{D(4e^{4B}+2e^{2B})},
\end{aligned}$$

where the last inequality holds due to the assumption of $t \geq 8(2e^{2B}+1)\sqrt{2DT\log\frac{T}{\rho}}$. □

A.2 Proof of Lemma 7

PROOF. By the definition of MLE loss function, we have

$$L_{\mathcal{S}_t}(c) = -\frac{1}{t}\sum_{i=1}^{t}\sum_{j=1}^{m_i}\log\left(\frac{\exp(c_{\sigma_i(j)})}{\sum_{k=j}^{D}\exp(c_{\sigma_i(k)})}\right),$$

which gives the gradient of the loss function with respect to c as

$$\begin{aligned}
\nabla L_{\mathcal{S}_t}(c) &= -\frac{1}{t}\sum_{i=1}^{t}\sum_{j=1}^{m_i}\frac{\partial\log\big(\frac{\exp(c_{\sigma_i(j)})}{\sum_{k=j}^{D}\exp(c_{\sigma_i(k)})}\big)}{\partial c}\\
&= -\frac{1}{t}\sum_{i=1}^{t}\sum_{j=1}^{m_i}\sum_{j'=j}^{D}\left(\frac{\exp(c_{\sigma_i(j')})}{\sum_{k=j}^{D}\exp(c_{\sigma_i(k)})}\right)(e_{\sigma_i(j')} - e_{\sigma_i(j)})
\end{aligned}$$

Define

$$V_i^{(jj')} = \begin{cases} \dfrac{\exp(\overline{c}^0_{\sigma_i(j')})}{\sum\limits_{k=j}^{D}\exp(\overline{c}^0_{\sigma_i(k)})}, & \text{if } \sigma_i(j) < \sigma_i(j') \\ -\dfrac{\exp(\overline{c}^0_{\sigma_i(j)})}{\sum\limits_{k=j}^{D}\exp(\overline{c}^0_{\sigma_i(k)})}, & \text{otherwise} \end{cases}$$

By [40], we have:

$$\mathbb{E}[V_i^{(jj')}] = 0 \quad \text{and} \quad \nabla L_{\mathcal{S}_t}(\overline{c}^0) = \frac{1}{t}\sum_{i=1}^{t}\sum_{j=1}^{m_i}\sum_{j'=j+1}^{D}V_i^{(jj')}(e_j - e_{j'})$$

Note:

$$\left\|\sum_{j'=j+1}^{D}V_i^{(jj')}(e_j - e_{j'})\right\| \leq \left\|\sum_{j'=j+1}^{D}V_i^{(jj')}e_j\right\| + \left\|\sum_{j'=j+1}^{D}V_i^{(jj')}e_{j'}\right\| = 2$$

And considering t interactions and m_j rounds of repeatedly selecting the most preferred objective from remaining objectives within each step $i \in [T]$, we see that $\nabla L_{\mathcal{S}_t}(\bar{c}^0)$ is the value of a discrete-time martingale at time $\sum_{i=1}^t m_i$, such that the martingale has initial value 0 and increments with norm bounded by 2. By Lemma 14 [20], we have:

$$\mathbb{P}\left[\left\|\nabla L_{\mathcal{S}_t}(\bar{c}^0)\right\|_2 \geq \sqrt{16t\bar{m}_t \log \tfrac{t}{\rho}}\right] \leq 2e^2 \exp\left(-\frac{16t\bar{m}_t \log \frac{t}{\rho}}{8\sum_{i=1}^t m_i}\right) = 2e^2\left(\frac{\rho}{t}\right)^2. \tag{10}$$

Summing over t from 1 to T, we have:

$$\sum_{t=1}^T 2e^2\left(\frac{\rho}{t}\right)^2 \leq 2e^2\rho^2 \sum_{t=1}^T \frac{1}{t^2} \leq \frac{e^2\pi^2\rho^2}{3}.$$

Thus, with probability at least $1 - \frac{e^2\pi^2\rho^2}{3}$, for any $t \in [T]$, we have:

$$\left\|\nabla L_{\mathcal{H}_t}(\bar{c}^0)\right\|_2 \leq \sqrt{16t\bar{m} \log \tfrac{t}{\rho}} \quad \text{holds.}$$

□

B Proof of Theorem 1

Proof. We proof the Theorem 1 using a simple toy instance with 2 arms and 2 objectives.

Let us consider an instance with two arms, where arm-1 has fixed reward $\boldsymbol{r}_1 = [0, 1]^\top$, arm-2 has fixed reward $\boldsymbol{r}_2 = [0.7, 0.7]^\top$. There two preference vectors $\boldsymbol{c}_1 = [0.1, 0.3]^\top$ and $\boldsymbol{c}_2 = [0.4, 0.6]^\top$. Apparently, under preference $\boldsymbol{c}_1$, arm-1 is the optimal arm, while for preference $\boldsymbol{c}_2$, arm-2 is the optimal.

Assume there exists a QE-only preference-aware algorithm $\mathcal{A}$ achieving sub-linear regret under preference $\boldsymbol{c}_1$. By Lemma 13, we have the expected number of pulls of arm-2 (suboptimal) given preference $\boldsymbol{c}_t$ is

$$\mathbb{E}_{\boldsymbol{c}_1}[N_{2,T}] = \sum_{t \in [T]} \mathbb{P}_{\pi_t^{\mathcal{A}} \mid \boldsymbol{c}_1}(a_t = 2 \mid \boldsymbol{c}_1) = o(T).$$

Due to the shift-invariant property of the Plackett–Luce (PL) model, any additive shift to the preference vector does not affect the induced ranking distribution. Specifically, since $\boldsymbol{c}_2 = \boldsymbol{c}_1 + 0.3 \cdot \mathbf{1}$, both $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$ induce the same distribution over rankings. Consequently, a QE-based estimator that relies solely on ranking data cannot distinguish between $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$, yielding the same mean-centered estimate in both cases. As a result, for any algorithm $\mathcal{A}$ interacting with a user whose true preference is $\boldsymbol{c}_2$, the probability of selecting any given arm remains identical to that under $\boldsymbol{c}_1$:

$$\mathbb{E}_{\boldsymbol{c}_2}[N_{2,T}] = \sum_{t=1}^T \mathbb{P}_{\pi_t^{\mathcal{A}} \mid \boldsymbol{c}_2}(a_t = 2) = \sum_{t=1}^T \mathbb{P}_{\pi_t^{\mathcal{A}} \mid \boldsymbol{c}_1}(a_t = 2) = \mathbb{E}_{\boldsymbol{c}_1}[N_{2,T}] = o(T).$$

However, recall that under preference $\boldsymbol{c}_2$, arm-2 is the optimal arm, which implies that the regret of $\mathcal{A}$ under preference $\boldsymbol{c}_2$ would be at least $\Omega(T)$, i.e.,

$$R_{\boldsymbol{c}_2}(T) = 1 \cdot \mathbb{E}_{\boldsymbol{c}_2}[N_{1,T}] = 1 \cdot (T - \mathbb{E}_{\boldsymbol{c}_2}[N_{2,T}]) > T - o(T) = \Omega(T).$$

□

C Proof of Lemma 2

Proof. For any $i \in [K], d \in [D], t > 0$, by Hoeffding's Inequality (Lemma 15), we have:

$$\mathbb{P}\left(|\hat{\boldsymbol{r}}_{i,t}(d) - \boldsymbol{\mu}_i(d)| > \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\right) \leq 2\exp\left(\frac{-2N_{i,t}^2 \log(t/\rho)}{N_{i,t}\sum_{\ell=1}^{N_{i,t}}(1-0)^2}\right) = 2\exp\left(-2\log(t/\rho)\right) = 2\left(\frac{\rho}{t}\right)^2, \tag{11}$$

Thus for any $i \in [K]$ and $t > 0$, with at least probability $1 - 2D\left(\frac{\rho}{t}\right)^2$, we have

$$\hat{\boldsymbol{r}}_{i,t} - \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1} \preceq \boldsymbol{\mu}_i \preceq \hat{\boldsymbol{r}}_{i,t} + \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1}.$$

And thus for any $n \in [N]$, we can derive

$$
\begin{aligned}
&\langle \hat{\boldsymbol{c}}_t^{n(\mathrm{BF})}, \hat{\boldsymbol{r}}_{i,t} \rangle + B_{i,t}^{n(r)} + B_{i,t}^{n(c)} - \langle \overline{\boldsymbol{c}}^n, \boldsymbol{\mu}_i \rangle \\
&\quad \geq \langle \hat{\boldsymbol{c}}_t^{n(\mathrm{BF})}, \hat{\boldsymbol{r}}_{i,t} \rangle + B_{i,t}^{n(r)} + B_{i,t}^{n(c)} - \left\langle \overline{\boldsymbol{c}}^n, \hat{\boldsymbol{r}}_{i,t} + \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1} \right\rangle \\
&\quad = \left\langle \hat{\boldsymbol{c}}_t^{n(\mathrm{BF})}, \hat{\boldsymbol{r}}_{i,t} + \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1} \right\rangle + B_{i,t}^{n(c)} - \left\langle \overline{\boldsymbol{c}}^n, \hat{\boldsymbol{r}}_{i,t} + \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1} \right\rangle \\
&\quad = \left\langle \hat{\boldsymbol{c}}_t^{n(\mathrm{BF})} - \overline{\boldsymbol{c}}^n, \hat{\boldsymbol{r}}_{i,t} + \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1} \right\rangle + B_{i,t}^{n(c)} \qquad (12) \\
&\quad \underset{(a)}{\geq} -\|\hat{\boldsymbol{c}}_t^{n(\mathrm{BF})} - \overline{\boldsymbol{c}}_t^n\|_{V_t^n} \cdot \left\| \hat{\boldsymbol{r}}_{i,t} + \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1} \right\|_{(V_t^n)^{-1}} + B_{i,t}^{n(c)} \\
&\quad \underset{(b)}{\geq} -\beta_t^n \cdot \left\| \hat{\boldsymbol{r}}_{i,t} + \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1} \right\|_{(V_t^n)^{-1}} + B_{i,t}^{n(c)} = 0, \\
&\qquad\qquad (\text{with probability at least } 1 - 2D\,(\rho/t)^2)
\end{aligned}
$$

where (a) holds by Cauchy-Schwarz inequality, (b) holds due to β_t^n as the radius of confidence region for preference estimation $\hat{\boldsymbol{c}}_t^{n(\mathrm{BF})}$. By the convergence of sum of reciprocals of squares that $\sum_{t=1}^{\infty} t^{-2} = \frac{\pi^2}{6}$, we have with probability larger than $1 - D\rho^2\pi^2/6$, above result holds for all t, completing the proof. □

D Proof of Lemma 3

Lemma 3 (Detailed Version). *For any user $n \in [N]$, any $\rho \in (0,1)$ and any sufficiently large samples $t \geq 8(2e^{2B}+1)\sqrt{2DT\log(\frac{T}{\rho})}$, with probability at least $1 - \frac{(1+D+2e^2)\rho^2\pi^2}{6}$, the estimated preference of Eq.4 in Algorithm 1 has the following upper-confidence bound*

$$
\|\hat{c}_t^{n(BF)} - \overline{c}^n\|_{V_t} \leq D(32e^{4B} + 16e^{2B})\sqrt{\frac{(1-\alpha)\log(t/\rho)}{\overline{m}_t t}} + B\sqrt{2(1-\alpha)\lambda} + \alpha\frac{BD}{2}\sqrt{D\log\left(\frac{1+\alpha D(t-1)\,(\lambda+1)}{\rho}\right) + \log\left(\frac{1+\lambda}{\lambda\rho}\right)}
$$

Proof. For notational simplicity, we fix a user n and omit the user index n as superscript. By Eq. 4, we have closed-form solution for preference estimation for bandit policy as:

$$
\hat{c}_t^{(\mathrm{BF})} = V_t^{-1}\left(\alpha\sum_{\tau=1}^{t-1} g_{a_\tau} r_{a_\tau,\tau} + (1-\alpha)U_\lambda \hat{c}_t^{(\mathrm{QE})}\right)
$$

Rearranging the equation yields

$$
\begin{aligned}
\hat{c}_t^{(\mathrm{BF})} &= V_t^{-1}\left(\alpha\sum_{\tau=1}^{t-1} g_{a_\tau} r_{a_\tau,\tau} + (1-\alpha)U_\lambda \hat{c}_t^{(\mathrm{QE})}\right) \\
&= V_t^{-1}\left(\alpha\sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top(\overline{c} + \xi_\tau) + (1-\alpha)U_\lambda \hat{c}_t^{(\mathrm{QE})}\right) \\
&= V_t^{-1}\left(\left(\alpha\sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top + (1-\alpha)U_\lambda\right)\overline{c} - (1-\alpha)U_\lambda\overline{c} + \alpha\sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top \xi_\tau + (1-\alpha)U_\lambda \hat{c}_t^{(\mathrm{QE})}\right) \\
&= \overline{c} - (1-\alpha)V_t^{-1}U_\lambda\overline{c} + \alpha V_t^{-1}\sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top \xi_\tau + (1-\alpha)V_t^{-1}U_\lambda \hat{c}_t^{(\mathrm{QE})},
\end{aligned}
$$

which implies that

$$
\|\hat{c}_t^{(\mathrm{BF})} - \overline{c}\|_{V_t} = \left\|(1-\alpha)V_t^{-1}U_\lambda(\hat{c}_t^{(\mathrm{QE})} - \overline{c}) + \alpha V_t^{-1}\sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top \xi_\tau\right\|_{V_t}
$$

Since V_t is PSD, we have:

$$
\|\hat{c}_t^{(\mathrm{BF})} - \overline{c}\|_{V_t} \leq (1-\alpha)\underbrace{\left\|V_t^{-1}U_\lambda(\hat{c}_t^{(\mathrm{QE})} - \overline{c})\right\|_{V_t}}_{A} + \alpha\underbrace{\left\|V_t^{-1}\sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top \xi_\tau\right\|_{V_t}}_{B} \qquad (13)
$$

For **Term A**, we have:

$$\begin{aligned} A &= \|V_t^{-1}U_\lambda(\hat{c}_t^{(\mathrm{QE})}-\overline{c})\|_{V_t} = \|U_\lambda(\hat{c}_t^{(\mathrm{QE})}-\overline{c})\|_{V_t}^{-1} \\ &\underset{(a)}{\leq} \frac{1}{\sqrt{1-\alpha}}\|U_\lambda(\hat{c}_t^{(\mathrm{QE})}-\overline{c})\|_{U_\lambda^{-1}} = \frac{1}{\sqrt{1-\alpha}}\|(\hat{c}_t^{(\mathrm{QE})}-\overline{c})\|_{U_\lambda}, \end{aligned} \tag{14}$$

where (a) holds since $V_t^{-1} \preceq V_{t-1}^{-1} \preceq V_1^{-1} = ((1-\alpha)U_\lambda)^{-1}$. For $\|(\hat{c}_t^{(\mathrm{QE})}-\overline{c})\|_{U_\lambda}$ we have

$$\begin{aligned} \left\|\hat{c}_t^{(\mathrm{QE})}-\overline{c}\right\|_{U_\lambda} &= \sqrt{(\hat{c}_t^{(\mathrm{QE})}-\overline{c})^\top U_\lambda(\hat{c}_t^{(\mathrm{QE})}-\overline{c})} \\ &= \sqrt{(\hat{c}_t^{(\mathrm{QE})}-\overline{c})^\top (U+\lambda I)(\hat{c}_t^{(\mathrm{QE})}-\overline{c})} \\ &= \sqrt{\underbrace{(\hat{c}_t^{(\mathrm{QE})}-\overline{c})^\top U(\hat{c}_t^{(\mathrm{QE})}-\overline{c})}_{C} + \lambda \underbrace{(\hat{c}_t^{(\mathrm{QE})}-\overline{c})^\top(\hat{c}_t^{(\mathrm{QE})}-\overline{c})}_{D}} \end{aligned}$$

For **Term C**, we expand:

$$\begin{aligned} (\hat{c}_t^{(\mathrm{QE})}-\overline{c})^\top U(\hat{c}_t^{(\mathrm{QE})}-\overline{c}) &= \left(\hat{c}_t^{(\mathrm{QE})} - \left(\overline{c}^0 + \tfrac{\|\overline{c}\|_1}{D}\mathbf{1}\right)\right)^\top U \left(\hat{c}_t^{(\mathrm{QE})} - \left(\overline{c}^0 + \tfrac{\|\overline{c}\|_1}{D}\mathbf{1}\right)\right) \\ &= (\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0)^\top U(\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0) + \left(\tfrac{\|\overline{c}\|_1}{D}\right)^2 \mathbf{1}^\top U\mathbf{1} - \tfrac{2\|\overline{c}\|_1}{D}\mathbf{1}^\top U(\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0) \end{aligned}$$

Note $U = I - \frac{1}{D}\mathbf{1}\mathbf{1}^\top$, so:

$$\mathbf{1}^\top U = \mathbf{1}^\top - \tfrac{1}{D}\mathbf{1}^\top\mathbf{1}^\top = 0, \quad U\mathbf{1} = 0 \Rightarrow \text{(all-ones projection vanishes)}$$

This gives that,

$$(\hat{c}_t^{(\mathrm{QE})}-\overline{c})^\top U(\hat{c}_t^{(\mathrm{QE})}-\overline{c}) = (\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0)^\top U(\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0) \quad \text{(orthogonal to} \mathbf{1}\text{)}.$$

For **Term D**, we have:

$$\begin{aligned} \lambda(\hat{c}_t^{(\mathrm{QE})}-\overline{c})^\top(\hat{c}_t^{(\mathrm{QE})}-\overline{c}) &= \lambda\left(\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0-\tfrac{\|\overline{c}\|_1}{D}\mathbf{1}\right)^\top\left(\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0-\tfrac{\|\overline{c}\|_1}{D}\mathbf{1}\right) \\ &\leq \lambda\left(\tfrac{\|\overline{c}\|_1}{D}\mathbf{1}+\tfrac{\|\overline{c}\|_1}{D}\mathbf{1}\right)^\top\left(\tfrac{\|\overline{c}\|_1}{D}\mathbf{1}+\tfrac{\|\overline{c}\|_1}{D}\mathbf{1}\right) \\ &= 2\lambda\cdot\tfrac{\|\overline{c}\|_1^2}{D^2}\cdot\mathbf{1}^\top\mathbf{1} \\ &= 2\lambda\cdot\tfrac{\|\overline{c}\|_1^2}{D^2}\cdot D = 2\lambda\cdot\tfrac{\|\overline{c}\|_1^2}{D} \end{aligned}$$

Combining together we have

$$\begin{aligned} \left\|\hat{c}_t^{(\mathrm{QE})}-\overline{c}\right\|_{U_\lambda} &= \sqrt{(\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0)^\top U(\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0) + \lambda\frac{\|\overline{c}\|_1^2}{D}} \\ &\leq \sqrt{(\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0)^\top U(\hat{c}_t^{(\mathrm{QE})}-\overline{c}^0)} + B\sqrt{2\lambda D} \\ &\leq D(32e^{4b}+16e^{2b})\sqrt{\frac{\log\frac{t}{\rho}}{\overline{m}_t t}} + B\sqrt{2\lambda D}. \\ &\qquad \text{(with probability} \geq 1-\frac{(D+2e^2)\pi^2\rho^2}{6} \text{ for all } t \text{ by Lemma 1.)} \end{aligned}$$

Plugging back Eq 14 gives:

$$A \leq \frac{1}{\sqrt{1-\alpha}}\left\|\hat{c}_t^{(\mathrm{QE})}-\overline{c}\right\|_{U_\lambda} \leq \frac{1}{\sqrt{1-\alpha}}\left(D(32e^{4b}+16e^{2b})\sqrt{\frac{\log\frac{t}{\rho}}{\overline{m}_t t}} + B\sqrt{2\lambda D}\right) \tag{15}$$

For **term B**, we first show that that $r_{a_\tau,\tau}^\top\xi_\tau$ is conditionally sub-Gaussian.

Let $\boldsymbol{a}_{1:t} = \{\boldsymbol{a}_\iota\}_{\iota=1}^t$ be the sequence of historical pulled actions within t steps, $\boldsymbol{g}_{1:t} = \{\boldsymbol{g}_{a_\iota,\iota}\}_{\iota=1}^t$ and $\boldsymbol{r}_{1:t} = \{\boldsymbol{r}_{a_\iota,\iota}\}_{\iota=1}^t$ be the sequences of historical overall scores and reward vectors within t steps respectively, and define the σ algebra $\mathcal{F}_{t-1} = \sigma(\boldsymbol{a}_{1:t-1}, \boldsymbol{g}_{1:t-1}, \boldsymbol{r}_{1:t})$. Note that for any $t \geq 1$,

$$\begin{aligned} \mathbb{E}[r_{a_\tau,\tau}^\top\xi_\tau \mid \mathcal{F}_{t-1}] &= \mathbb{E}[c_t^\top r_{a_t,t} \mid \mathcal{F}_{t-1}] - \mathbb{E}[\overline{c}^\top r_{a_t,t} \mid \mathcal{F}_{t-1}] \\ &\underset{(a)}{=} \overline{\boldsymbol{c}}^{t-1}\boldsymbol{r}_{a_t,t} - \overline{\boldsymbol{c}}^{t-1}\boldsymbol{r}_{a_t,t} = 0, \end{aligned}$$

where (a) holds since $\boldsymbol{c}_t$ is independent of $\mathcal{F}_{t-1}$ and the conditional expectation fact that $\mathbb{E}(X \mid \mathcal{F}) = X$ if $X \in \mathcal{F}$. Furthermore, by assumption of DB-bounded overall-reward, $-DB \le r_{a_\tau,\tau}^\top \xi_\tau \le DB$ holds almost surely, and hence we can conclude that $r_{a_\tau,\tau}^\top \xi_\tau$ is conditionally DB-sub-Gaussian. Also, since $\boldsymbol{r}_{a_t,t}$ is $\mathcal{F}_{t-1}$-measurable, define $r'_{a_\tau,\tau} = \sqrt{\alpha} r_{a_\tau,\tau}$, by Lemma 16, we have with probability at least $1-\rho$:

$$\left\| \sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top \xi_\tau \right\|_{V_t^{-1}} = \frac{1}{\sqrt{\alpha}} \left\| \sum_{\tau=1}^{t-1} r'_{a_\tau,\tau} r'^\top_{a_\tau,\tau} \xi_\tau \right\|_{V_t^{-1}} \le \frac{DB}{\sqrt{\alpha}} \sqrt{2 \log\left(\frac{\det(V_t)^{1/2} \det(V_1)^{-1/2}}{\rho} \right)}. \tag{16}$$

Note $V_1 = (1-\alpha) U_\lambda = (1-\alpha)\left(I - \frac{1}{D}\mathbf{1}\mathbf{1}^\top + \lambda I\right)$, it can be easily verified that:

$$\det(U_\lambda) = \lambda(1+\lambda)^{D-1} \Rightarrow (\text{eigenvalues: } \lambda \text{ with multiplicity } 1, \quad 1+\lambda \text{ with multiplicity } D-1).$$

Since

$$V_t = (1-\alpha)U_\lambda + \sum_{\tau=1}^{t-1} r'_{a_\tau,\tau} r'^\top_{a_\tau,\tau} \preceq (1-\alpha)(1+\lambda) I + \sum_{\tau=1}^{t-1} r'_{a_\tau,\tau} r'^\top_{a_\tau,\tau}, \quad \text{with } \|r'_{a_\tau,\tau}\|_2 \le \sqrt{\alpha D},$$

we have

$$\det(V_t) \le ((1-\alpha)(1+\lambda))^D \cdot \left(1 + \tfrac{\alpha D(t-1)}{1+\lambda}\right)^D.$$

And note $\det(V_1) = (1-\alpha)^D \det(U_\lambda) = (1-\alpha)^D \lambda (1+\lambda)^{D-1}$, we have

$$\frac{\det(V_t)}{\det(V_1)} = \frac{1+\lambda}{\lambda} \cdot \left(1 + \tfrac{\alpha D(t-1)}{1+\lambda}\right)^D.$$

Plugging back to Eq. 16 gives (with probability at least $1-\rho$):

$$\left\| \sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top \xi_\tau \right\|_{V_t^{-1}} \le BD\sqrt{D \log\left(\frac{1+\lambda}{\lambda\rho} \cdot \frac{1+\alpha D(t-1)/(1+\lambda)}{\rho} \right)}.$$

Combine A and B together, we have

$$\begin{aligned}
\left\| \hat{c}_t^{(\mathrm{BF})} - \bar{c} \right\|_{V_t} &\le (1-\alpha) \left\| V_t^{-1} U_\lambda (\hat{c}_t^{(\mathrm{QE})} - \bar{c}) \right\|_{V_t} + \alpha \left\| V_t^{-1} \sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top \xi_\tau \right\|_{V_t} \\
&\le (1-\alpha) \left(\frac{D(32e^{4b} + 16e^{2b})}{\sqrt{1-\alpha}} \cdot \sqrt{\frac{\log \frac{t}{\rho}}{\overline{m}_t t}} + B\sqrt{\frac{2\lambda}{1-\alpha}} \right) \\
&\quad + \alpha BD \sqrt{D \log\left(\frac{1+\alpha D(t-1)/(1+\lambda)}{\rho} \right) + \log\left(\tfrac{1+\lambda}{\lambda\rho}\right)},
\end{aligned}$$

thus completing the proof. ◻

E Proof of Theorem 2

Before the detailed proofs, we first restate Theorem 2 with a detained version and show how the optimal choices of balance-weight α and regularization λ are derived.

Theorem 2 (Detailed Version)**.** *For any user $n \in [N]$, any $\rho \in (0,1)$ and any sufficiently large samples $t \ge 8(2e^{2B}+1)\sqrt{2DT\log(\frac{T}{\rho})}$, with probability at least $1 - \frac{(1+D+2e^2)\rho^2\pi^2}{6}$, the estimated preference of Eq.4 in Algorithm 1 has the following upper-confidence bound For any user $n \in [N]$, by setting β_t as the UCB of $\hat{\mathbf{c}}_t^{n(BF)}$ in Lemma 3, with probability at least $1 - \frac{(2+D+2e^2)\rho^2\pi^2}{6}$, we have the following regret bound*

$$R^n(T) \le \Psi^n_{\hat{c}}(T) + \Psi^n_{\hat{r}}(T) + M, \quad \text{where}$$

$$M = \max\{\tfrac{2D\alpha}{\nu_{r\downarrow}^2}\sqrt{KT\log(\tfrac{T}{\rho})}, \tfrac{1}{\alpha}\log(\tfrac{T}{\rho}), 8(2e^{2B}+1)\sqrt{2DT\log(\tfrac{T}{\rho})}\},$$

$$\Psi^n_{\hat{c}}(T) = O\Big(\big(\sqrt{\tfrac{D^3}{\overline{m}_T}} + B\sqrt{\tfrac{\lambda}{\alpha}} + BD^{\frac{3}{2}}\sqrt{\alpha\log(\tfrac{T}{\rho}) + \alpha\log(\tfrac{1+\lambda}{\lambda\rho})}\big)\sqrt{T\log(\tfrac{\alpha T}{\lambda})}\Big),$$

$$\Psi^n_{\hat{r}}(T) = O\Big(\sqrt{\tfrac{KD^3}{\overline{m}_T\lambda}}\log^2 T + \big(BD + \tfrac{\alpha BD^{\frac{3}{2}}}{\sqrt{\lambda}}\sqrt{\log(\tfrac{T}{\rho}) + \log(\tfrac{1+\lambda}{\lambda\rho})}\big)\sqrt{KT\log(\tfrac{T}{\rho})}\Big).$$

Remark 1. *With above result, we can optimize the choices of balance-weight α and regularization λ to minimize the user-specific regret. Specifically, the bound reveals the main multipliers of $\sqrt{T \log T}$. By setting $\alpha = \lambda = O(\log^{-1}(T/\rho))$, we can guarantee all such multipliers become $O(1)$, thus achieving a near optimal regret*

$$R^n(T) = O(\sqrt{T \log T}).$$

Compared to the previous PRUCB [9] in preference-aware MO-MAB, our proactive QE-based framework and algorithm achieve achieves much better performance by reducing a multiplicative $\sqrt{\log T}$ term.

Remark 2 (Final Regret). *By summing the optimized user-specific regret in Remark 1 over all users $n \in [N]$, we have the final regret:*

$$R(T) = \textstyle\sum_{n=1}^{N} R^n(T) = O\left(N\sqrt{T \log T}\right),$$

assuming the same algorithmic parameters $\alpha = \lambda = O(\log^{-1}(T/\rho))$ are used for each user.

Before the proof, we present some essential observations that will be used in the proof.

Claim 1. *By Hoeffding's Inequality (Lemma 15), with probability at least $1 - \frac{KD\rho^2\pi^2}{3}$, we have*

$$|\boldsymbol{\mu}_i(d) - \hat{\boldsymbol{r}}_{i,t}(d)| \leq \sqrt{\frac{\log(\frac{t}{\rho})}{N_{i,t}}}$$

holds for all $i \in [K], d \in [D], t \in [T]$.

Claim 2. *Since $(\alpha \boldsymbol{r}_{i,\ell}\boldsymbol{r}_{i,\ell}^{\top})(m,n) \in [0,\alpha], \forall (m,n) \in [D]$ and by Hoeffding's Inequality (Lemma 15), with probability at least $1 - \frac{KD(D-1)\rho^2\pi^2}{12}$, we have*

$$\mathbb{E}\left[\sum_{\ell \in \mathcal{T}_{i,t-1}} \alpha \boldsymbol{r}_{i,\ell}\boldsymbol{r}_{i,\ell}^{\top}\right](m,n) - \sum_{\ell \in \mathcal{T}_{i,t-1}} \left(\alpha \boldsymbol{r}_{i,\ell}\boldsymbol{r}_{i,\ell}^{\top}\right)(m,n) \leq \alpha\sqrt{N_{i,t}\log\left(\frac{t}{\rho}\right)}$$

holds $\forall i \in [K], \forall m \in [D], \forall n \in [m, D]$.

Proof of Theorem 2. Here we focus on the user-specific regret $R^n(T)$. For notational simplicity, we fix a user n and omit the user index n as superscript. Let M be an arbitrary positive integer, we can express the user-specific regret $R^n(T)$ in a truncated form with respect to M as follows:

$$R(T)^n = \sum_{t=1}^{T} \text{regret}_t \leq O(M) + \sum_{t=M+1}^{T} \text{regret}_t, \tag{17}$$

where regret_t denotes the instantaneous regret of algorithm at step $t \in [T]$, and the last inequality holds since the fact that the instantaneous regret is bounded.

Next, we analyze the instantaneous regret over the truncated time horizon $[M+1, T]$. By Claim 1, we have

$$\boldsymbol{\mu}_{a_t^*} \preceq \hat{\boldsymbol{r}}_{a_t^*,t} + \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1}, \text{ and } \boldsymbol{\mu}_{a_t} \succeq \hat{\boldsymbol{r}}_{a_t,t} - \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1}. \tag{18}$$

By the definition of regret and fact above, we can derive the upper-bound of expected instantaneous regret as follows:

$$\begin{aligned}
\text{regret}_t &= \overline{\boldsymbol{c}}^{\top}\boldsymbol{\mu}_{a_t^*} - \overline{\boldsymbol{c}}^{\top}\boldsymbol{\mu}_{a_t} \\
&\underset{(a)}{\leq} \hat{\boldsymbol{c}}^{\top}\hat{\boldsymbol{r}}_{a_t^*,t} + B^r_{a_t^*,t} + B^c_{a_t^*,t} - \overline{\boldsymbol{c}}^{\top}\boldsymbol{\mu}_{a_t} \\
&\underset{(b)}{\leq} \hat{\boldsymbol{c}}^{\top}\hat{\boldsymbol{r}}_{a_t^*,t} + B^r_{a_t^*,t} + B^c_{a_t^*,t} - \overline{\boldsymbol{c}}^{\top}\left(\hat{\boldsymbol{r}}_{a_t,t} + \sqrt{\frac{\log(t/\rho)}{N_{a,t}}}\mathbf{1}\right) + 2\|\overline{\boldsymbol{c}}\|_1\sqrt{\frac{\log(t/\rho)}{N_{a,t}}} \\
&\underset{(c)}{\leq} \hat{\boldsymbol{c}}^{\top}\hat{\boldsymbol{r}}_{a_t,t} + B^r_{a_t,t} + B^c_{a_t,t} - \overline{\boldsymbol{c}}^{\top}\left(\hat{\boldsymbol{r}}_{a_t,t} + \sqrt{\frac{\log(t/\rho)}{N_{a,t}}}\mathbf{1}\right) + 2\|\overline{\boldsymbol{c}}\|_1\sqrt{\frac{\log(t/\rho)}{N_{a,t}}} \\
&= (\hat{\boldsymbol{c}} - \overline{\boldsymbol{c}})^{\top}\left(\hat{\boldsymbol{r}}_{a_t,t} + \sqrt{\frac{\log(t/\rho)}{N_{a,t}}}\mathbf{1}\right) + B^c_{a_t,t} + 2\|\overline{\boldsymbol{c}}\|_1\sqrt{\frac{\log(t/\rho)}{N_{a,t}}} \\
&\underset{(d)}{\leq} \|\hat{\boldsymbol{c}}_t - \overline{\boldsymbol{c}}_t\|_{\boldsymbol{V}_{t-1}} \cdot \left\|\hat{\boldsymbol{r}}_{i,t} + \sqrt{\frac{\log(t/\rho)}{N_{a,t}}}\mathbf{1}\right\|_{\boldsymbol{V}_{t-1}^{-1}} + B^c_{a_t,t} + 2\|\overline{\boldsymbol{c}}\|_1\sqrt{\frac{\log(t/\rho)}{N_{a,t}}} \\
&\underset{(e)}{\leq} \min\left(2B^c_{a_t,t}, BD\right) + 2\|\overline{\boldsymbol{c}}\|_1\sqrt{\frac{\log(t/\rho)}{N_{a,t}}},
\end{aligned}$$

where (a) followed by Lemma 2, (b) followed by Eq. 18, (c) holds by the definition of optimization policy for arm selection, (d) holds by Cauchy-Schwarz inequality, (e) followed by the definition of $B^c_{a_t,t}$, and the fact that the instantaneous regret is at most BD. Above result implies

$$
\begin{aligned}
R(T) &\le O(M) + \sum_{t=M+1}^{T} \text{regret}_t \\
&\le O(M) + \sum_{t=M+1}^{T} \left(\text{regret}_t^{\tilde{c}} + \text{regret}_t^{\tilde{r}}\right) \\
&\le O(M) + \underbrace{\sum_{t=M+1}^{T} \min\left(2B^c_{a_t,t}, BD\right)}_{R^{\tilde{c}}_{M+1:T}} + \underbrace{\sum_{t=M+1}^{T} 2\|\bar{\boldsymbol{c}}\|_1 \sqrt{\frac{\log(t/\alpha)}{N_{a,t}}}}_{R^{\tilde{r}}_{M+1:T}},
\end{aligned}
\tag{19}
$$

Next we analyze two components of $R^{\tilde{c}}_{M+1:T}$ and $R^{\tilde{r}}_{M+1:T}$ separately.

Step-1. (Upper-Bound over $R^{\tilde{c}}_{M+1:T}$)

We first present two useful lemmas:

Lemma 8. *For any action sequence of $a_1, ..., a_T$ and any $M \in [0, T]$, we have*

$$
\det\left(\mathbb{E}[\boldsymbol{V}_T]\right) \ge \det\left(\mathbb{E}[\boldsymbol{V}_M]\right) \prod_{t=M}^{T-1} \left(1 + \alpha \boldsymbol{\mu}_{a_t}^\top \mathbb{E}[\boldsymbol{V}_t]^{-1} \boldsymbol{\mu}_{a_t}\right).
$$

Please see Appendix E.1 for the detailed proof of Lemma 8.

Lemma 9. *For any action sequence of $a_1, ..., a_T$, any weight $\alpha > 0$, and any $M \in [1, T]$, we have*

$$
\log\left(\frac{\det\left(\mathbb{E}[\boldsymbol{V}_T]\right)}{\det\left(\mathbb{E}[\boldsymbol{V}_M]\right)}\right) \le D \log\left(1 + \frac{\alpha}{\lambda(1-\alpha)}(T - M)\right).
$$

Please see Appendix E.2 for the detailed proof of Lemma 9.

Under Claim 1, for any $t > 1$ and $i \in [K]$, we have

$$
\left\|\hat{\boldsymbol{r}}_{i,t} + \sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1} - \boldsymbol{\mu}_i\right\|_2 \le \left\|2\sqrt{\frac{\log(t/\rho)}{N_{i,t}}}\mathbf{1}\right\|_2 = 2\sqrt{D\frac{\log(t/\rho)}{N_{i,t}}} = 2\sqrt{D}\gamma_{i,t}.
$$

Consequently,

$$
\begin{aligned}
\left\|\hat{\boldsymbol{r}}_{i,t} + \sqrt{\log(t/\rho)/N_{i,t}}\mathbf{1}\right\|_{\boldsymbol{V}_t^{-1}} - \|\boldsymbol{\mu}_i\|_{\boldsymbol{V}_t^{-1}} &\le \left\|\hat{\boldsymbol{r}}_{i,t} + \sqrt{\log(t/\rho)/N_{i,t}}\mathbf{1} - \boldsymbol{\mu}_i\right\|_{\boldsymbol{V}_t^{-1}} \\
&\le \frac{1}{\sqrt{\lambda}}\left\|\hat{\boldsymbol{r}}_{i,t} + \sqrt{\log(t/\rho)/N_{i,t}}\mathbf{1} - \boldsymbol{\mu}_i\right\|_2 \le 2\sqrt{\frac{D}{\lambda}}\gamma_{i,t},
\end{aligned}
$$

$$
\Longrightarrow \left\|\hat{\boldsymbol{r}}_{i,t} + \sqrt{\log(t/\rho)/N_{i,t}}\mathbf{1}\right\|_{\boldsymbol{V}_t^{-1}} \le \|\boldsymbol{\mu}_i\|_{\boldsymbol{V}_t^{-1}} + 2\sqrt{\frac{D}{\lambda}}\gamma_{i,t}. \tag{20}
$$

Define $M = \left\lfloor \min\left\{t' \mid tv^2_{r\downarrow} + \lambda \ge 2D\alpha\sqrt{Kt\log\frac{t}{\rho}}, \forall t \ge t'\right\}\right\rfloor$. Please note that for $v^2_{r\downarrow} > 0$, we have $\lim_{t\to\infty} \frac{2D\alpha\sqrt{Kt\log\frac{t}{\rho}}}{v^2_{r\downarrow}t} = \lim_{t\to\infty} C_1\sqrt{\frac{\log(t)-C_2}{t}} = 0$ since as t increase, $\sqrt{\log t}$ grows much slowly compared to $\sqrt{t}$. Hence for sufficiently large t', the inequality $tv^2_{r\downarrow} + \lambda \ge 2D\alpha\sqrt{Kt\log\frac{t}{\rho}}, \forall t \ge t'$ holds, which implies that such an M does indeed exist. By Lemma 18, for any $t \in [M+1, T]$, we have

$$
\begin{aligned}
R^{\tilde{c}}_{M+1:T} &= \sum_{t=M+1}^{T} \min\left(2B^{c}_{a_t,t}, BD\right) \\
&= \sum_{t=M+1}^{T} \min\left(2\beta_t \left\|\hat{\boldsymbol{r}}_{a_t,t} + \sqrt{\log(t/\rho)/N_{a_t,t}}\mathbf{1}\right\|_{\boldsymbol{V}_t^{-1}}, BD\right) \\
&\underset{(a)}{\leq} \sum_{t=M+1}^{T} \min\left(2\beta_t \left(\|\boldsymbol{\mu}_{a_t}\|_{\boldsymbol{V}_t^{-1}} + \frac{1}{\sqrt{\lambda}}\gamma_{a_t,t}\right), BD\right) \\
&\underset{(b)}{\leq} \sum_{t=M+1}^{T} \min\left(2\beta_t \left(\sqrt{2}\|\boldsymbol{\mu}_{a_t}\|_{\mathbb{E}[\boldsymbol{V}_t]^{-1}} + \frac{1}{\sqrt{\lambda}}\gamma_{a_t,t}\right), BD\right) \\
&\underset{(c)}{\leq} 2\sqrt{2}\underbrace{\sum_{t=M+1}^{T} \min\left(\beta_t \|\boldsymbol{\mu}_{a_t}\|_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}, \frac{BD}{2\sqrt{2}}\right)}_{\mathrm{Reg}^{(1)}} + 4\underbrace{\sum_{t=M+1}^{T} \beta_t \sqrt{\frac{D}{\lambda}}\gamma_{a_t,t}}_{\mathrm{Reg}^{(2)}},
\end{aligned}
\tag{21}
$$

Step-1.1. (Upper-Bound over $\mathrm{Reg}_t^{(1)}$)

Plugging the $\beta_t = D(32e^{4B} + 16e^{2B})\sqrt{\frac{(1-\alpha)\log(t/\rho)}{\overline{m}_t t}} + B\sqrt{2(1-\alpha)\lambda} + \alpha\frac{BD}{2}\sqrt{D\log\left(\frac{1+\alpha D(t-1)/(\lambda+1)}{\rho}\right) + \log(\frac{1+\lambda}{\lambda\rho})}$ (defined in Lemma 3), we have

$$
\begin{aligned}
\mathrm{Reg}^{(1)} \leq{}& D(32e^{4B} + 16e^{2B})\sqrt{\frac{(1-\alpha)}{\overline{m}_t}} \sum_{t=M+1}^{T} \min\left(\sqrt{\frac{\log(t/\rho)}{t}}\|\boldsymbol{\mu}_{a_t}\|_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}, \frac{BD}{2\sqrt{2}}\right) \\
&+ \left(B\sqrt{2\lambda(1-\alpha)} + \alpha\frac{BD}{2}\sqrt{D\log\left(\frac{1+\alpha DT/(\lambda+1)}{\rho}\right) + \log(\frac{1+\lambda}{\lambda\rho})}\right) \sum_{t=M+1}^{T} \min\left(\|\boldsymbol{\mu}_{a_t}\|_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}, \frac{BD}{2\sqrt{2}}\right)
\end{aligned}
$$

By Cauchy–Schwarz inequality, we have

$$
\begin{aligned}
\mathrm{Reg}^{(1)} \leq{}& D(32e^{4B} + 16e^{2B})\sqrt{\frac{(1-\alpha)}{\overline{m}_t}} \sqrt{\sum_{t=M+1}^{T} 1 \cdot \underbrace{\sum_{t=M+1}^{T} \min\left(\frac{\log(t/\rho)}{t}\|\boldsymbol{\mu}_{a_t}\|^2_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}, \frac{B^2D^2}{8}\right)}_{I_1}} \\
&+ \left(B\sqrt{2\lambda(1-\alpha)} + \alpha\frac{BD}{2}\sqrt{D\log\left(\frac{1+\alpha DT/(\lambda+1)}{\rho}\right) + \log(\frac{1+\lambda}{\lambda\rho})}\right) \sqrt{\sum_{t=M+1}^{T} 1 \cdot \underbrace{\sum_{t=M+1}^{T} \min\left(\|\boldsymbol{\mu}_{a_t}\|^2_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}, \frac{B^2D^2}{8}\right)}_{I_2}}.
\end{aligned}
\tag{22}
$$

For $\boldsymbol{I}_1$, note for sufficiently large enough iterations $t \geq \frac{\log(T/\rho)}{\alpha}$, we have $\frac{\log(T/\rho)}{t} \leq \alpha$, thus

$$
\begin{aligned}
\sum_{t=M+1}^{T} \min\left(\frac{\log(t/\rho)}{t}\|\boldsymbol{\mu}_{a_t}\|^2_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}, \frac{B^2D^2}{8}\right) &\leq \sum_{t=M+1}^{T} \min\left(\alpha\|\boldsymbol{\mu}_{a_t}\|^2_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}, \frac{B^2D^2}{8}\right) \\
&\underset{(a)}{\leq} c_0 \sum_{t=1}^{T} \log\left(1 + \alpha\|\boldsymbol{\mu}_{a_t}\|^2_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}\right) \\
&\underset{(b)}{\leq} c_0 D \log\left(1 + \frac{\alpha}{\lambda(1-\alpha)}(T-M)\right),
\end{aligned}
\tag{23}
$$

where (a) holds by using the inequality $c_0\log(1+x) \geq \frac{B^2D^2}{8\log(1+\frac{B^2D^2}{8})}\log(1+x) \geq x$, with some constant c_0 for $x \in [0, \frac{B^2D^2}{8}]$, last inequality (b) holds by chaining the results of Lemma 8 and Lemma 9, and plugging back to the result above.

For $\boldsymbol{I}_2$, similarly, we have:

$$
\begin{aligned}
\sum_{t=M+1}^{T} \min\left(\|\boldsymbol{\mu}_{a_t}\|^2_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}, \frac{B^2D^2}{8}\right) &\leq \frac{1}{\alpha}\sum_{t=M+1}^{T} \min\left(\alpha\|\boldsymbol{\mu}_{a_t}\|^2_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}, \frac{B^2D^2}{8}\right) \\
&\leq \frac{c_0}{\alpha}\sum_{t=1}^{T}\log\left(1+\alpha\|\boldsymbol{\mu}_{a_t}\|^2_{\mathbb{E}[\boldsymbol{V}_t]^{-1}}\right) \\
&\leq \frac{c_0 D}{\alpha}\log\left(1+\frac{\alpha}{\lambda(1-\alpha)}(T-M)\right).
\end{aligned}
\tag{24}
$$

Combining Eq. 23 and Eq. 24 yields for $t \geq \frac{\log(T/\rho)}{\alpha}$,

$$
\begin{aligned}
\mathrm{Reg}^{(1)} &\leq D(32e^{4B}+16e^{2B})\sqrt{\frac{c_0D(1-\alpha)}{\overline{m}_t}}\sqrt{(T-M)\log\left(1+\frac{\alpha}{\lambda(1-\alpha)}(T-M)\right)} \\
&+\left(B\sqrt{\frac{2\lambda(1-\alpha)c_0D}{\alpha}}+\frac{BD}{2}\sqrt{\alpha c_0 D}\sqrt{D\log\left(\frac{1+\alpha DT/(\lambda+1)}{\rho}\right)+\log\left(\frac{1+\lambda}{\lambda\rho}\right)}\right)\sqrt{(T-M)\log\left(1+\frac{\alpha}{\lambda(1-\alpha)}(T-M)\right)}.
\end{aligned}
\tag{25}
$$

Step-1.2. (Upper-Bound over $\mathrm{Reg}_t^{(2)}$)

Plugging the $\beta_t = D(32e^{4B}+16e^{2B})\sqrt{\frac{(1-\alpha)\log(t/\rho)}{\overline{m}_t t}} + B\sqrt{2(1-\alpha)\lambda} + \alpha\frac{BD}{2}\sqrt{D\log\left(\frac{1+\alpha D(t-1)/(\lambda+1)}{\rho}\right)+\log\left(\frac{1+\lambda}{\lambda\rho}\right)}$ (defined in Lemma 3), we have

$$
\begin{aligned}
\mathrm{Reg}^{(2)} &\leq D^2(32e^{4B}+16e^{2B})\sqrt{\frac{(1-\alpha)}{m_T^{\downarrow}\lambda}}\sum_{t=M+1}^{T}\sqrt{\frac{\log^2(t/\rho)}{tN_{a_t,t}}} \\
&+\left(B\sqrt{2\lambda(1-\alpha)}+\alpha\frac{BD}{2}\sqrt{D\log\left(\frac{1+\alpha DT/(\lambda+1)}{\rho}\right)+\log\left(\frac{1+\lambda}{\lambda\rho}\right)}\right)\sqrt{\frac{D}{\lambda}}\sum_{t=M+1}^{T}\sqrt{\frac{\log(t/\rho)}{N_{a_t,t}}}.
\end{aligned}
$$

By Lemma 19 and Lemma 20, we have

$$
\begin{aligned}
\mathrm{Reg}^{(2)} &\leq O\left(D(32e^{4B}+16e^{2B})\sqrt{\frac{D(1-\alpha)}{\lambda}}\sqrt{K\log^2\left(\frac{T}{\rho}\right)}\right) \\
&+O\left(\left(B\sqrt{2D(1-\alpha)}+\alpha\frac{BD}{2}\sqrt{\frac{D^2}{\lambda}\log\left(\frac{1+\alpha DT/(\lambda+1)}{\rho}\right)+\frac{D}{\lambda}\log\left(\frac{1+\lambda}{\lambda\rho}\right)}\right)\sqrt{KT\log\left(\frac{T}{\rho}\right)}\right).
\end{aligned}
\tag{26}
$$

Step-2. (Upper-Bound over $R^{\tilde{r}}_{M+1:T}$)

$$
\sum_{t=M+1}^{T} 2\|\overline{\boldsymbol{c}}\|_1\sqrt{\frac{\log(t/\alpha)}{N_{a,t}}} \leq 2BD\sum_{t=M+1}^{T}\sqrt{\frac{\log(t/\alpha)}{N_{a,t}}} \leq 4BD\sqrt{K(T-M)\log\left(\frac{T}{\rho}\right)}
\tag{27}
$$

where the final inequality holds by Lemma 19.

Step-3. (Combine the Bounds) Summing up the results of Eq. 25, Eq. 26 and Eq. 27 together complete the proof. □

E.1 Proof of Lemma 8

Proof of Lemma 8. Let $\boldsymbol{r}'_{a_t,t} = \sqrt{\alpha}\boldsymbol{r}_{a_t,t}$, $\boldsymbol{\mu}'_{a_t} = \sqrt{\alpha}\boldsymbol{\mu}_{a_t}$. For $\boldsymbol{V}_t$ and $\mathbb{E}[\boldsymbol{V}_t]$, by definition,

$$
\boldsymbol{V}_{t+1} = \boldsymbol{V}_t + \alpha\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^{\top} \quad \text{and} \quad \boldsymbol{V}_1 = \lambda\boldsymbol{I}+\boldsymbol{U}
$$

$$
\mathbb{E}[\boldsymbol{V}_{t+1}] = \mathbb{E}[\boldsymbol{V}_t + \alpha\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^{\top}] = \mathbb{E}[\boldsymbol{V}_t + \boldsymbol{r}'_{a_t,t}\boldsymbol{r}'^{\top}_{a_t,t}] = \mathbb{E}[\boldsymbol{V}_t] + \boldsymbol{\mu}'_{a_t}\boldsymbol{\mu}'^{\top}_{a_t} + \alpha\Sigma_{r,a_t}.
$$

Since $\mathbb{E}[\boldsymbol{V}_t]$ is symmetric and positive definite, we have

$$
\begin{aligned}
\det\left(\mathbb{E}[\boldsymbol{V}_{t+1}]\right) &= \det\left(\mathbb{E}[\boldsymbol{V}_t] + \boldsymbol{\mu}'_{a_t}\boldsymbol{\mu}'^{\top}_{a_t} + \alpha\Sigma_{r,a_t}\right) \\
&= \det\left(\mathbb{E}[\boldsymbol{V}_t]^{\frac{1}{2}}\left(\boldsymbol{I}+\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\left(\boldsymbol{\mu}'_{a_t}\boldsymbol{\mu}'^{\top}_{a_t}+\alpha\Sigma_{r,a_t}\right)\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\right)\mathbb{E}[\boldsymbol{V}_{t-1}]^{\frac{1}{2}}\right) \\
&= \det\left(\mathbb{E}[\boldsymbol{V}_{t-1}]\right)\det\left(\boldsymbol{I}+\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\left(\boldsymbol{\mu}'_{a_t}\boldsymbol{\mu}'^{\top}_{a_t}+\alpha\Sigma_{r,a_t}\right)\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\right) \\
&\underset{(a)}{\geq} \det\left(\mathbb{E}[\boldsymbol{V}_t]\right)\left(\det\left(\boldsymbol{I}+\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\boldsymbol{\mu}'_{a_t}\boldsymbol{\mu}'^{\top}_{a_t}\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\right)+\det\left(\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\alpha\Sigma_{r,a_t}\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\right)\right)
\end{aligned}
\tag{28}
$$

where (a) holds since $\left(\boldsymbol{I} + \mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\boldsymbol{\mu}'_{a_t}\boldsymbol{\mu}'^{\top}_{a_t}\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\right)$ and $\left(\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\alpha\Sigma_{r,a_t}\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\right)$ are positive definite and applying Lemma 17 yields the result.

Let $\mathbb{E}[\boldsymbol{V}_{t-1}]^{-\frac{1}{2}}\boldsymbol{\mu}'_{a_t} = \boldsymbol{v}_t$, and we observe that

$$\left(\boldsymbol{I} + \boldsymbol{v}_t\boldsymbol{v}_t^\top\right)\boldsymbol{v}_t = \boldsymbol{v}_t + \boldsymbol{v}_t\left(\boldsymbol{v}_t^\top\boldsymbol{v}_t\right) = \left(1 + \boldsymbol{v}_t^\top\boldsymbol{v}\right)\boldsymbol{v}_t.$$

Hence, $1 + \boldsymbol{v}_t^\top\boldsymbol{v}$ is an eigenvalue of $\boldsymbol{I} + \boldsymbol{v}_t\boldsymbol{v}_t^\top$. And since $\boldsymbol{v}_t\boldsymbol{v}_t^\top$ is a rank-1 matrix, all other eigenvalue of $\boldsymbol{I} + \boldsymbol{v}_t\boldsymbol{v}_t^\top$ equal to 1, implying

$$\begin{aligned}\det\left(\boldsymbol{I} + \mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\boldsymbol{\mu}'_{a_t}\boldsymbol{\mu}'^{\top}_{a_t}\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\right) &= \det\left(\boldsymbol{I} + \boldsymbol{v}_t\boldsymbol{v}_t^\top\right)\\ &= 1 + \boldsymbol{v}_t\boldsymbol{v}_t^\top\\ &= 1 + \left(\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\boldsymbol{\mu}'_{a_t}\right)^\top\left(\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\boldsymbol{\mu}'_{a_t}\right)\\ &= 1 + \boldsymbol{\mu}'^{\top}_{a_t}\mathbb{E}[\boldsymbol{V}_t]^{-1}\boldsymbol{\mu}'_{a_t}.\end{aligned} \tag{29}$$

Combining Eq. 28 and Eq. 29, we have

$$\begin{aligned}\det\left(\mathbb{E}[\boldsymbol{V}_{t+1}]\right) &\geq \det\left(\mathbb{E}[\boldsymbol{V}_t]\right)\left(1 + \boldsymbol{\mu}'^{\top}_{a_t}\mathbb{E}[\boldsymbol{V}_t]^{-1}\boldsymbol{\mu}'_{a_t} + \det\left(\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\Sigma_{r,a_t}\mathbb{E}[\boldsymbol{V}_t]^{-\frac{1}{2}}\right)\right)\\ &\geq \det\left(\mathbb{E}[\boldsymbol{V}_t]\right)\left(1 + \boldsymbol{\mu}'^{\top}_{a_t}\mathbb{E}[\boldsymbol{V}_t]^{-1}\boldsymbol{\mu}'_{a_t}\right)\end{aligned}$$

The solution of Lemma 8 follows from induction. □

E.2 Proof of Lemma 9

Proof. By the definition of $\boldsymbol{V}_t$, we have

$$\begin{aligned}\log\left(\frac{\det\left(\mathbb{E}[\boldsymbol{V}_T]\right)}{\det\left(\mathbb{E}[\boldsymbol{V}_M]\right)}\right) &= \log\left(\det\left(\frac{\mathbb{E}[\boldsymbol{V}_M] + \sum_{t=M}^{T-1}\mathbb{E}[\alpha\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top]}{\mathbb{E}[\boldsymbol{V}_M]}\right)\right)\\ &\underset{(a)}{\leq} \log\left(\det\left(\left((1-\alpha)\boldsymbol{U}_\lambda + \sum_{t=M}^{T-1}\mathbb{E}[\alpha\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top]\right)\frac{1}{1-\alpha}\boldsymbol{U}_\lambda^{-1}\right)\right)\\ &= \log\left(\det\left(\boldsymbol{I} + \frac{\alpha}{1-\alpha}\sum_{t=M}^{T-1}\mathbb{E}[\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top]\boldsymbol{U}_\lambda^{-1}\right)\right),\end{aligned} \tag{30}$$

where (a) holds since $\det(\mathbb{E}[\boldsymbol{V}_M]) \geq \det(\mathbb{E}[\boldsymbol{V}_1]) = \boldsymbol{U}_\lambda$. By Sherman–Morrison formula, we have:

$$\begin{aligned}U_\lambda^{-1} &= \left(\lambda I + I - \tfrac{1}{D}\mathbf{1}\mathbf{1}^\top\right)^{-1} = \left((\lambda+1)I - \tfrac{1}{D}\mathbf{1}\mathbf{1}^\top\right)^{-1}\\ &= \frac{1}{\lambda+1}I + \frac{\frac{1}{D(\lambda+1)^2}\mathbf{1}\mathbf{1}^\top}{1 - \frac{1}{D}\cdot\frac{1}{\lambda+1}\cdot\mathbf{1}^\top\mathbf{1}} = \frac{1}{\lambda+1}I + \frac{\frac{1}{D(\lambda+1)^2}\mathbf{1}\mathbf{1}^\top}{1 - \frac{1}{\lambda+1}}\end{aligned} \tag{31}$$

Let $\xi_1, ..., \xi_D$ denote the eigenvalues of $\sum_{t=M}^{T-1}\mathbb{E}[\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top]\boldsymbol{U}_\lambda^{-1}$, and note:

$$\begin{aligned}\sum_{d=1}^{D}\xi_d &= \text{Trace}\left(\sum_{t=M}^{T-1}\mathbb{E}[\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top]\boldsymbol{U}_\lambda^{-1}\right)\\ &\underset{(a)}{=} \frac{\sum_{t=M}^{T-1}\mathbb{E}\left[\text{Trace}\left(\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top\right)\right]}{\lambda+1} + \frac{\sum_{t=M}^{T-1}\text{Trace}\left(\mathbb{E}[\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top]\mathbf{1}\mathbf{1}^\top\right)}{D(\lambda+1)\lambda}\\ &\underset{(b)}{=} \frac{\sum_{t=M}^{T-1}\mathbb{E}\left[\|\boldsymbol{r}_{a_t,t}\|_2^2\right]}{\lambda+1} + \frac{\sum_{t=M}^{T-1}\mathbb{E}[\|\boldsymbol{r}_{a_t,t}\|_1^2]}{D(\lambda+1)\lambda}\\ &\underset{(c)}{\leq} \frac{D(T-M)}{\lambda+1} + \frac{D^2(T-M)}{D(\lambda+1)\lambda}\\ &= \frac{D}{\lambda}(T-M).\end{aligned} \tag{32}$$

where (a) holds by Eq. 31, (c) holds since we assume $\boldsymbol{r}_t \leq [1]^D$, (b) holds since

$$\begin{aligned}\text{Trace}\left(\mathbb{E}[\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top]\mathbf{1}\mathbf{1}^\top\right) &= \mathbb{E}[\text{Trace}\left(\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top\mathbf{1}\mathbf{1}^\top\right)] = \mathbb{E}[(\boldsymbol{r}_{a_t,t}^\top\mathbf{1})\text{Trace}\left(\boldsymbol{r}_{a_t,t}\mathbf{1}^\top\right)]\\ &= \mathbb{E}[(\boldsymbol{r}_{a_t,t}^\top\mathbf{1})^2] = \mathbb{E}[\|\boldsymbol{r}_{a_t,t}\|_1^2].\end{aligned}$$

Combining Eq. 30 and Eq. 32 implies

$$
\begin{aligned}
\log\left(\frac{\det\left(\mathbb{E}[\boldsymbol{V}_T]\right)}{\det\left(\mathbb{E}[\boldsymbol{V}_M]\right)}\right) &\leq \log\left(\det\Big(\boldsymbol{I}+\frac{\alpha}{1-\alpha}\sum_{t=M}^{T-1}\mathbb{E}[\boldsymbol{r}_{a_t,t}\boldsymbol{r}_{a_t,t}^\top]\boldsymbol{U}_\lambda^{-1}\Big)\right)\\
&= \log\left(\prod_{i=1}^{D}\left(1+\frac{\alpha}{1-\alpha}\xi_i\right)\right)\\
&= D\log\left(\prod_{i=1}^{D}\left(1+\frac{\alpha}{1-\alpha}\xi_i\right)\right)^{\frac{1}{D}}\\
&\underset{(a)}{\leq} D\log\left(\frac{1}{D}\sum_{i=1}^{D}\left(1+\frac{\alpha}{1-\alpha}\xi_i\right)\right)\\
&\underset{(b)}{\leq} D\log\left(1+\frac{\alpha D(T-M)}{D\lambda(1-\alpha)}\right),
\end{aligned}
$$

where (a) follows from the inequality of arithmetic and geometric means, and (b) follows from Eq. 32. □

F Proof of Theorem 3

Proof. We prove the result with a constructed instance.

Construct instances with a pair of two different zero-mean preference vectors:

$$
c=\begin{cases}\frac{\Delta}{2}, & i\leq\lfloor\frac{D}{2}\rfloor \quad \text{(upper list)}\\ -\frac{\Delta}{2}, & \lfloor\frac{D}{2}\rfloor<i\leq D \text{ (lower list)}\end{cases},\quad c'=\begin{cases}-\frac{\Delta}{2}, & i\leq\lfloor\frac{D}{2}\rfloor \quad \text{(upper list)}\\ \frac{\Delta}{2}, & \lfloor\frac{D}{2}\rfloor<i\leq D \text{ (lower list)}\end{cases}
$$

Let $P=\frac{e^{\Delta/2}}{e^{\Delta/2}+e^{-\Delta/2}}=\sigma(\Delta)$ denote the probability of objectives with preference value $\frac{\delta}{2}$ preferred over objectives with preference of $-\frac{\delta}{2}$. Under ϵ-corruption model with arbitrary flipping, we can easily verify that the corrupted pairwise comparison probability with largest distribution shift is caused by entries swaps between upper and lower lists, resulting the pairwise comparison probability between objectives in upper list and lower list as:

$$
\widetilde{P}_{ij}^{(c)}=(1-\epsilon)P+\epsilon(1-P)=(1-2\epsilon)P+\epsilon
$$

$$
\widetilde{P}_{ij}^{(c')}=(1-\epsilon)(1-P)+\epsilon P=(2\epsilon-1)P-\epsilon+1
$$

We leverage the ranking-breaking strategy that reduces the full ranking σ into independent pairwise comparisons, where each compared objective being preferred follows a Bernoulli distribution shown above. Let $\sigma=(i_1\succ i_2\succ\cdots\succ i_D)$ be a full ranking over D items. We model the likelihood as:

$$
P(\sigma\mid c)=\prod_{(i\succ j)\in\sigma}\mathrm{Ber}(1;\widetilde{P}_{ij}^{(c)})
$$

where each pairwise outcome Y_{ij} is drawn independently as:

$$
Y_{ij}\sim\mathrm{Ber}(\widetilde{P}_{ij}^{(c)})
$$

Thus, the full-data distribution becomes a product of independent Bernoulli comparisons:

$$
P(\{Y_{ij}\}\mid c)=\prod_{(i,j)\in S}\mathrm{Ber}(Y_{ij};\widetilde{P}_{ij}^{(c)})
$$

Now consider set S of all (i,j) pairs where $i\in[\lfloor\frac{D}{2}\rfloor]$, $j\in[\lfloor\frac{D}{2}\rfloor+1,D]$ with $|S|=\frac{D^2}{4}$. Then the KL divergence between two corrupted models with underlying c and c' is given as

$$
\mathrm{KL}(P^{(c)}\|P^{(c')})=\sum_{(i,j)\in S}\mathrm{KL}(\mathrm{Ber}(\widetilde{P}_{ij}^{(c)})\|\mathrm{Ber}(\widetilde{P}_{ij}^{(c')}))
$$

Define $\delta=\frac{1}{2}\left|\widetilde{P}_{ij}^{(c)}-\widetilde{P}_{ij}^{(c')}\right|=(1-2\epsilon)(P-\frac{1}{2})=(1-2\epsilon)\tanh(\Delta/2)$ for $\epsilon<\frac{1}{2}$.

Apply standard bound for Bernoulli KL divergence (since $\widetilde{P}_{ij}^{(c)}+\widetilde{P}_{ij}^{(c')}=1$):

$$
\mathrm{KL}\left(\mathrm{Ber}(\widetilde{P}_{ij}^{(c)})\middle\|\mathrm{Ber}(\widetilde{P}_{ij}^{(c')})\right)\leq 4\delta^2=4(1-2\epsilon)^2\tanh^2(\Delta/2)
$$

Thus we have,

$$
\mathrm{KL}(P^{(c)}\|P^{(c')})\leq|S|\cdot 4(1-2\epsilon)^2\tanh^2(\Delta/2)=D^2(1-2\epsilon)^2\tanh^2(\Delta/2).
$$

Using Pinsker's inequality, for L ranking samples with $\epsilon < \frac{1}{2}$, we have:

$$\mathrm{TV}((P^{(c)})^L \| (P^{(c')})^L) = \sqrt{\frac{L}{2}\mathrm{KL}(P^{(c)} \| P^{(c')})} \leq \sqrt{\frac{L}{2}} D(1-2\epsilon)\tanh(\frac{\Delta}{2})$$

Solving for $\mathrm{TV} = \frac{1}{2}$ gives,

$$\tanh(\frac{\Delta}{2}) = \frac{1}{\sqrt{2L}D(1-2\epsilon)} \implies \Delta = c_0 \cdot 2\sqrt{2}\tanh^{-1}(\frac{2}{(1-2\epsilon)D\sqrt{L}}),$$

with some constant $c_0 > 0$. Finally, let $\hat{c}_L$ be any estimator with L samples, by Le Cam's lemma we have:

$$\begin{aligned}\inf_{\hat{\boldsymbol{c}}_L} \sup_{\overline{\boldsymbol{c}}^0 \in \{c, c'\}} \mathbb{E}\big[\|\hat{\boldsymbol{c}}_L - \overline{\boldsymbol{c}}^0\|_2\big] &\geq \frac{\|c - c'\|_2}{2}(1 - \mathrm{TV}((P^{(c)})^L \| (P^{(c')})^L)) \\ &= \frac{\sqrt{D}\Delta}{4} = c_0\sqrt{\frac{D}{2}}\tanh^{-1}\left(\frac{2}{(1-2\epsilon)D\sqrt{L}}\right),\end{aligned}$$

with the valid region of $\frac{2}{(1-2\epsilon)D\sqrt{L}} < 1$, yields $\epsilon \leq \frac{1}{2} - O(\frac{1}{D\sqrt{L}})$. Additionally, we further have $\|\hat{\boldsymbol{c}}_L - \overline{\boldsymbol{c}}^0\|_2$ be bounded by $2B\sqrt{D}$ due to the bounded region of c. For $\epsilon > \frac{1}{2}$, it can be easily verify that $P^{(c)}$ and $P^{(c')}$ are distinguishable, leading to a constant estimator error gap. □

G Proof of Lemma 4

Proof. Note for any number of sample t, we define the error components as $[\delta_i]_t = [\delta_1, \ldots, \delta_t] \in \mathbb{R}^{D\times t}$, where $\delta_i \in \mathbb{R}^D$. Recall we have the solution for the estimator

$$(\hat{c}_t^{(\mathrm{QE})}, [\hat{\delta}_i]_t) = \arg\min_{c\in\Theta_c^0, [\hat{\delta}_i]_t} \hat{\mathcal{L}}_{\tilde{S}_t}(c, \delta) + \lambda\sum_{j=1}^t \|\delta_j\|_2$$

where

$$\hat{\mathcal{L}}_{\tilde{S}_t}(c, [\hat{\delta}_i]_t) = -\frac{1}{t}\sum_{i=1}^t\sum_{j=1}^{m_i}\log\left(\frac{\exp(c_{\sigma_i(j)} + \delta_{i,\sigma_i(j)})}{\sum_{k=d}^D \exp(c_{\sigma_i(k)} + \delta_{i,\sigma_i(k)})}\right)$$

Define $\Delta_c = \hat{c}^{(\mathrm{QE})} - \overline{c}^0$, $\Delta_\delta = [\hat{\delta}_i]_t - [\delta_i^*]_t \in \mathbb{R}^{D\times T}$ and $\Delta_\delta^{\mathrm{flat}} = \mathrm{vec}(\Delta_\delta) \in \mathbb{R}^{DT}$, where $\mathrm{vec}(\cdot)$ denotes column-major flattening. By the definition of the MLE estimator, we have:

$$\hat{\mathcal{L}}_{\tilde{S}_t}(c, [\hat{\delta}_i]_t) + \lambda\sum_{i=1}^t\|\hat{\delta}_i\|_2 \leq \hat{\mathcal{L}}_{\tilde{S}_t}(c, [\delta_i^*]_t) + \lambda\sum_{i=1}^t\|\delta_i^*\|_2$$

Due to strong convexity of $\hat{\mathcal{L}}_{\tilde{S}_t}$ (by Lemma 10), we can rewrite as:

$$\begin{aligned}\eta\sum_{i=1}^t\|\delta_i^*\|_2 - \eta\sum_{i=1}^t\|\hat{\delta}_i\|_2 &\geq \hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c}_t^{(QE)}, [\hat{\delta}_i]_t) - \hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0, [\delta_i^*]_t) \\ &= \hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c}_t^{(QE)}, [\delta_i^*]_t + \Delta_\delta) - \hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c}_t^{(QE)}, [\delta_i^*]_t) \\ &\quad + \hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0 + \Delta_c, [\delta_i^*]_t) - \hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0, [\delta_i^*]_t) \\ &\geq \nabla_c\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0, [\delta_i^*]_t)^\top\Delta_c + \nabla_\delta\hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c}_t^{(QE)}, [\delta_i^*]_t)^\top\Delta_\delta^{\mathrm{flat}} \\ &\quad + \frac{t\overline{m}_t}{D(4e^{4B} + 2e^B)}\|\Delta_c\|_2^2 + \frac{m_t^\downarrow}{D(4e^{4B} + 2e^B)}\|\Delta_\delta^{\mathrm{flat}}\|_2^2.\end{aligned}$$

Reorganizing above result gives

$$\begin{aligned}&\frac{t\overline{m}_t}{D(4e^{4B} + 2e^B)}\|\Delta_c\|_2^2 + \frac{m_t^\downarrow}{D(4e^{4B} + 2e^B)}\|\Delta_{\mathcal{S}}^{\mathrm{flat}}\|_2^2 \\ &\leq \eta\sum_{i=1}^t\|\hat{\delta}_i^*\|_2 - \eta\sum_{i=1}^t\|\hat{\delta}_i\|_2 - \nabla_c\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0, [\delta_i^*]_t)^\top\Delta_c - \nabla_{\mathcal{S}}\hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c}_t^{\mathrm{QE}}, [\delta_i^*]_t)^\top\Delta_\delta^{\mathrm{flat}}.\end{aligned}$$

Let $\mathcal{T}_t^c$ denote the set of time steps where ranking is corrupted within t samples, and $|\mathcal{T}_t^c| = N_t^c$, let $\Delta_{\delta+}^{\text{flat}} = \Delta_{\delta}^{\text{flat}}[\mathcal{T}_t^c]$ denote the gap of estimated δ over corrupted samples. By Lemma 11, with $\eta = \frac{2D+9p+2p^2D^2+3+6(D^2+t)}{27}$, we have:

$$\begin{aligned}
&\frac{t\overline{m}_t}{D(4e^{4B}+2e^{B})}\|\Delta_c\|_2^2 + \frac{m_t^{\downarrow}}{D(4e^{4B}+2e^{B})}\|\Delta_{\delta}^{\text{flat}}\|_2^2 \\
&\le 2\sqrt{N_t^c}\eta\|\Delta_{\delta}^{\text{flat}}\|_2 + \|\nabla_c\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\|_2\cdot\|\Delta_c\|_2 \\
&\le \left(2\sqrt{N_t^c}\eta + \sqrt{\frac{1}{t}}\|\nabla_c\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\|_2\right)\left(\|\Delta_{\delta+}^{\text{flat}}\|_2 + \sqrt{t}\|\Delta_c\|_2\right) \\
&\le \left(2\sqrt{N_t^c}\eta + \sqrt{16D\log\tfrac{t}{\rho}}\right)\sqrt{2(\|\Delta_{\delta+}^{\text{flat}}\|_2^2 + t\|\Delta_c\|_2^2)},
\end{aligned}$$

where the last inequality holds by Eq. 10 and Cauchy–Schwarz inequality. This implies

$$\begin{aligned}
\sqrt{2}D(4e^{4B}+2e^{B})\left(2\sqrt{N_t^c}\eta + \sqrt{16D\log\tfrac{1}{\rho}}\right) &\ge \frac{t\overline{m}_t\|\Delta_c\|_2^2 + m_t^{\downarrow}\|\Delta_{\delta}^{\text{flat}}\|_2^2}{\sqrt{t\|\Delta_c\|_2^2 + \|\Delta_{\delta+}^{\text{flat}}\|_2^2}} \\
&\ge \sqrt{m_t^{\downarrow}}\cdot\sqrt{t\|\Delta_c\|_2^2 + \|\Delta_{\delta+}^{\text{flat}}\|_2^2}.
\end{aligned}$$

Note that within t steps, by Chernoff bound on Bernoulli distribution (Lemma 21) of corruption $B(\varepsilon)$, the number of corruption time satisfies:

$$\mathbb{P}\left(N_t^c \ge \varepsilon t + 2\sqrt{\varepsilon t\log\left(\tfrac{t}{\rho}\right)}\right) \le \left(\tfrac{\rho}{t}\right)^2$$

$$\Rightarrow \quad N_t^c \ge \varepsilon t + 2\sqrt{\varepsilon t\log\left(\tfrac{t}{\rho}\right)} \quad \text{with high probability } (1-\tfrac{\rho^2}{t^2}).$$

Finally, for any $t \ge 8(2e^{2B}+1)\sqrt{2Dt\log\frac{1}{\xi}}$, by setting $\eta \ge \frac{2D+9D+(2D^2+6)\sqrt{D^2+3}}{27}$, we have:

$$\begin{aligned}
\|\Delta_c\|_2^2 + \|\Delta_{\delta+}^{\text{flat}}\|_2^2 &\le \frac{2D^2(4e^{4B}+2e^{B})^2}{tm_t^{\downarrow}}\left(2\eta\sqrt{\varepsilon t + 2\left(\varepsilon t\log(\tfrac{t}{\rho})\right)^{1/2}} + \sqrt{16D\log\tfrac{t}{\rho}}\right)^2 \\
&\le \frac{4D^2(4e^{4B}+2e^{B})^2}{m_t^{\downarrow}}\left(4\eta^2(\varepsilon + 2\sqrt{\frac{\varepsilon}{t}\log(\tfrac{t}{\rho})}) + 16D\log\left(\tfrac{t}{\rho}\right)\right)
\end{aligned}$$

holds with probability at least $1-(1+D+2e^2)\cdot\frac{\rho^2}{t^2}$, which complete the proof □

Lemma 10 (Strong Convexity). *The loss function $\hat{\mathcal{L}}_{\tilde{S}_t}$ is strongly convex in the following sense:*

(i) *$\hat{\mathcal{L}}_{\tilde{S}_t}$ is strongly convex with respect to c at $(\overline{c}^0,[\delta_i^*]_t)$:*

$$\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0+\Delta_c,[\delta_i^*]_t) - \hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t) - \nabla_c\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)^\top\Delta_c \ge \frac{t\overline{m}_t}{D(4e^{4B}+2e^{B})}\|\Delta_c\|_2^2$$

(ii) *$\hat{\mathcal{L}}_{\tilde{S}_t}$ is strongly convex with respect to δ at $(\hat{c}^{(QE)},[\delta_i^*]_t)$:*

$$\hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c}^{(QE)},[\delta_i^*]_t+\Delta_\delta) - \hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c}^{(QE)},[\delta_i^*]_t) - \nabla_\delta\hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c}^{(QE)},[\delta_i^*]_t)^\top\Delta_\delta^{flat} \ge \frac{m_t^{\downarrow}}{D(4e^{4B}+2e^{B})}\|\Delta_\delta^{flat}\|_2^2$$

The proof is provided in Appendix G.1.

Lemma 11. *Let*

$$\eta \ge \frac{2D+9p+2p^2(D^2+3)+6(D^2+3)}{27},$$

then the following inequality holds:

$$\begin{aligned}
&\lambda\sum_{i=1}^{t}\|\delta_i^*\|_2 - \lambda\sum_{i=1}^{t}\|\hat{\delta}_i\|_2 - \nabla_c\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)^\top\Delta_c - \nabla_\delta\hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c}^{(QE)_t},[\delta_i^*]_t)^\top\Delta_\delta^{flat} \\
&\qquad\le 2\sqrt{N_t^c}\eta\|\Delta_{\delta+}^{flat}\|_2 + \|\nabla_c\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\|_2\cdot\|\Delta_c\|_2.
\end{aligned} \tag{33}$$

The proof is provided in Appendix G.2.

G.1 Proof of Lemma 10

Proof. Without loss of generality, we fixed the total number of sample as t and omit the index of t for the simplicity.

(i) Strongly Convex w.r.t c at $(\bar{c}^0, [\delta_i^*]_t)$

We follow some results of QE preference estimator in uncorrupted case for proof. Specifically, let

$$p_k^{(i,j)} = \frac{\mathbb{1}_k^{(i,j)} \exp(c_k + \delta_{i,k})}{\sum_{k' \in S^{(i,j)}} \exp(c_{k'} + \delta_{i,k'})}$$

denote the probability for selection in $S^{i,j}$. The Hessian with respect to c can be written as:

$$H_c(c, [\delta]_t) = \sum_{i=1}^{t} \sum_{j=1}^{m_i} H^{(i,j)}, \quad \text{where}$$

$$H^{(i,j)} = \frac{1}{2} \sum_{j,j' \in S^{i,j}} (e_k - e_{k'})(e_k - e_{k'})^\top p_k^{(i,j)} p_{k'}^{(i,j)} = \text{diag}(p^{(i,j)}) - p^{(i,j)}(p^{(i,j)})^\top.$$

By Lemma 6, we have

$$\xi_2(H_c(c, [\delta]_t)) \geq \frac{t\overline{m}_t}{D(4e^{4B} + 2e^{2B})}.$$

Thus for any $u \in \Theta_c^0$, we have

$$u^\top H_c(c, [\delta]_t) u \geq \xi_2(H_c(c, [\delta]_t)) \|u\|_2^2 \geq \frac{t\overline{m}_t}{D(4e^{4B} + 2e^{2B})} \|u\|_2^2,$$

which implies the strong convexity of $\hat{\mathcal{L}}_{\tilde{S}_t}$ w.r.t c at $(\bar{c}^0, [\delta_i^*]_t)$.

(ii) Strongly Convex w.r.t δ at $(\hat{c}^{(\text{QE})}, [\delta_i^*]_t)$

Since each ranking sample σ_i is only dependent on the model of $PL(c + \delta_i)$, we have the gradient $\nabla_\delta \hat{\mathcal{L}}_{\tilde{S}_t}(c, [\delta_i^*]_t)$ splits *sample-by-sample*. In this case, the Hessian $H_\delta(c, [\delta]_t)$ is *block-diagonal*, i.e., $D \times D$ blocks for each ranking sample σ_i. And thus, we only need to check the convexity *sample-by-sample*.

Specifically, for each sample σ_i, the Hessian w.r.t δ_i is

$$H_{\delta_i}^{(i)}(c, [\delta]_t) = \sum_{j=1}^{m_i} H^{(i,j)}, \quad \text{where} \quad H^{(i,j)} = \text{diag}(p^{(i,j)}) - p^{(i,j)}(p^{(i,j)})^\top \in \mathbb{R}^{D \times D}.$$

And it gives the final Hessian w.r.t $[\delta_i]_t$ as:

$$H_\delta(c, [\delta]_t) = \begin{bmatrix} H_{\delta_1}^{(1)} & & & \\ & H_{\delta_1}^{(1)} & & \\ & & \ddots & \\ & & & H_{\delta_t}^{(t)} \end{bmatrix} \in \mathbb{R}^{Dt \times Dt}.$$

Recall that each δ_i is induced by swaps, for any $i \in [t]$, we have $\delta_i \perp \mathbf{1}$, and by Lemma 6, we have

$$\xi_2(H_{\delta_i}^{(i)}(c, [\delta]_t)) \geq \frac{m_i}{D(4e^{4B} + 2e^{2B})}. \tag{34}$$

Thus for any $\delta_i \in \Theta_\delta$ and $U = \begin{bmatrix} u_1 \\ \vdots \\ u_t \end{bmatrix} \in \mathbb{R}^{D \cdot t}$ with $u_i \in \mathbb{R}^D$ and $u_i \perp \mathbf{1}, \forall i \in [t]$, we have

$$u^\top H_{\delta_i}^{(i)} u \geq \xi_2(H_{\delta_i}^{(i)}) \|u\|_2^2 \geq \frac{m_i}{D((4e^{4B} + 2e^{2B})} \|u\|_2^2,$$

which implies the strong convexity of $H_{\delta_i}^{(i)}$, and for the final Hessian, we have

$$\begin{aligned} U^\top H_\delta(c, [\delta]_t) U &= [u_1^\top H_{\delta_1}^{(1)}, u_2^\top H_{\delta_2}^{(2)}, ..., u_t^\top H_{\delta_t}^{(t)}] U = \sum_{i=1}^{t} u_i H_{\delta_i}^{(i)} u_i \\ &\geq \sum_{i=1}^{t} \xi_2\left(H_{\delta_i}^{(i)}\right) \|u_i\|_2^2 \geq \min_{i \in [t]} \xi_2\left(H_{\delta_i}^{(i)}\right) \sum_{i=1}^{t} \|u_i\|_2^2 = \frac{m_t^\downarrow}{D(4e^{4B} + 2e^{2B})} \|U\|_2^2, \end{aligned}$$

where the last equality holds by Eq. 34 and $\sum_{i=1}^t \|u_i\|_2^2 = \|U\|_2^2$. Above result implies the strong convexity of $\hat{\mathcal{L}}_{\tilde{S}_t}$ w.r.t $[\delta_i]_t$ at $(\bar{c}^0, [\delta_i^*]_t)$. □

G.2 Proof of Lemma 11

We first present a useful lemma that will be used for proof.

Lemma 12. *For any ranking sample δ_i under any corrupted Plackett-Luce model* $\mathrm{PL}(\delta_i)$*, we have:*

$$\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta^*]_t)\right\|_2 \le \frac{2D^3+9p+2p^2\sqrt{D^2+3}+6\sqrt{D^2+3}}{27}$$

The proof is provided in Appendix G.3

Proof of Lemma 11. Let $\mathcal{T}_t^c$ denote the set of time steps where ranking is corrupted within t samples, and $|\mathcal{T}_t^c| = N_t^c$, let $\Delta^{\text{flat}}_{\delta+} = \Delta^{\text{flat}}_{\delta}[\mathcal{T}_t^c]$ denote the gap of estimated δ over corrupted samples. we have

$$\begin{aligned}
\sum_{i=1}^{t}\|\delta_i^*\|_2 - \sum_{i=1}^{t}\|\hat{\delta}_i\|_2 &= \sum_{i\in\mathcal{T}_t^c}\left(\|\delta_i^*\|_2-\|\hat{\delta}_i\|_2\right) + \sum_{i\in[t]\setminus\mathcal{T}_t^c}\left(\|\delta_i^*\|_2-\|\hat{\delta}_i\|_2\right)\\
&= \sum_{i\in\mathcal{T}_t^c}\left(\|\delta_i^*\|_2-\|\hat{\delta}_i\|_2\right) - \sum_{i\in[t]\setminus\mathcal{T}_t^c}\|\hat{\delta}_i\|_2\\
&\le \sum_{i\in\mathcal{T}_t^c}\|\delta_i^*-\hat{\delta}_i\|_2 - \sum_{i\in[t]\setminus\mathcal{T}_t^c}\|\hat{\delta}_i\|_2\\
&\le \sqrt{N_t^c}\Big(\sum_{i\in\mathcal{T}_t^c}\|\delta_i^*-\hat{\delta}_i\|_2^2\Big)^{1/2} - \sum_{i\in[t]\setminus\mathcal{T}_t^c}\|\hat{\delta}_i\|_2\\
&= \sqrt{N_t^c}\cdot\|\Delta^{\text{flat}}_{\delta+}\|_2 - \sum_{i\in[t]\setminus\mathcal{T}_t^c}\|\hat{\delta}_i\|_2
\end{aligned} \tag{35}$$

On the other hand, we have

$$\begin{aligned}
&\nabla_\delta\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)^\top\Delta_\delta\\
&\quad= \sum_{i\in\mathcal{T}_t^c}\left|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)^\top(\delta_i^*-\hat{\delta}_i)\right| + \sum_{i\in[t]\setminus\mathcal{T}_t^c}\left|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)^\top(\delta_i^*-\hat{\delta}_i)\right|\\
&\quad\ge -\sum_{i\in\mathcal{T}_t^c}\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2\cdot\left\|\delta_i^*-\hat{\delta}_i\right\|_2 - \sum_{i\in[t]\setminus\mathcal{T}_t^c}\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2\cdot\left\|\hat{\delta}_i\right\|_2\\
&\quad\ge -\max_{i\in\mathcal{T}_t^c}\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2\cdot\sum_{i\in\mathcal{T}_t^c}\left\|\delta_i^*-\hat{\delta}_i\right\|_2 - \max_{i\in[t]\setminus\mathcal{T}_t^c}\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2\cdot\sum_{i\in[t]\setminus\mathcal{T}_t^c}\left\|\hat{\delta}_i\right\|_2\\
&\quad\ge -\max_{i\in\mathcal{T}_t^c}\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2\cdot\sqrt{N_t^c}\cdot\left\|\Delta^{(\text{flat})}_{\delta_t}\right\|_2 - \max_{i\in[t]\setminus\mathcal{T}_t^c}\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2\cdot\sum_{i\in[t]\setminus\mathcal{T}_t^c}\left\|\hat{\delta}_i\right\|_2
\end{aligned} \tag{36}$$

Combining above results together yields

$$\begin{aligned}
&\eta\sum_{i=1}^{t}\|\delta_i^*\|_2 - \eta\sum_{i=1}^{t}\|\hat{\delta}_i\|_2 - \nabla_c\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)^\top\Delta_c - \nabla_\delta\hat{\mathcal{L}}_{\tilde{S}_t}(\hat{c},\delta^*)^\top\Delta^{\text{flat}}_\delta\\
&\quad\le \sqrt{N_t^c}\left(\max_{i\in\mathcal{T}_t^c}\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2+\eta\right)\left\|\Delta^{\text{flat}}_{\delta+}\right\|_2\\
&\quad+\left(\max_{i\in[t]\setminus\mathcal{T}_t^c}\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2-\eta\right)\sum_{i\in[t]\setminus\mathcal{T}_t^c}\|\hat{\delta}_i\|_2\\
&\quad+\left\|\nabla_c\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2\cdot\|\Delta_c\|_2
\end{aligned} \tag{37}$$

By Lemma 12, setting

$$\eta \ge \frac{2D^3+9p+2p^2\sqrt{D^2+3}+6\sqrt{D^2+3}}{27} \ge \max_{i\in[t]}\left\|\nabla^{(i)}_{\delta_i}\hat{\mathcal{L}}_{\tilde{S}_t}(\overline{c}^0,[\delta_i^*]_t)\right\|_2$$

completes the proof.

□

G.3 Proof of Lemma 12

Proof. For each sample δ_i, the gradient with respect to δ_i is:

$$\nabla^{(i)}_{\delta_i} \tilde{\mathcal{L}}_{\tilde{S}_t}(c, [\delta]_t) = -\sum_{j=1}^{m_i} \frac{2}{D} \left[\log \left(\frac{\exp(c_i(j) + \delta_i(j))}{\sum_{k=j}^{D} \exp(c_i(k) + \delta_i(k))} \right) \right].$$

Using the gradient of softmax log-likelihood, we get:

$$\nabla^{(i)}_{\delta_i} \tilde{\mathcal{L}}_{\tilde{S}_t}(c, [\delta]_t) = -\sum_{j=1}^{m_i} \sum_{i'=i+1}^{D} \left(\frac{\exp(c_i(j) + \delta_i(j))}{\sum_{k=i}^{D} \exp(c_i(k) + \delta_i(k))} (e_i(j) - e_i(j')) \right).$$

From this, we can bound each coordinate:

$$\left| \nabla^{(i)}_{\delta_i} \tilde{\mathcal{L}}_{\tilde{S}_t}(c, [\delta]_t)(j) \right| \leq \begin{cases} 1 - \frac{D}{\sum_{k=j}^{D} \exp(c_i(k)+\delta_i(k))} \exp(c_i(j) + \delta_i(j)) \leq 1, & j \leq m_i \\ \sum_{j'=1}^{m_i} \frac{\exp(c_i(j)+\delta_i(j))}{\sum_{k=j}^{D} \exp(c_i(k)+\delta_i(k))} \leq m_i, & j > m_i \end{cases}$$

This implies the total ℓ_2 norm is bounded by:

$$\left\| \nabla^{(i)}_{\delta_i} \tilde{\mathcal{L}}_{\tilde{S}_t}(c, [\delta]_t) \right\|_2 \leq \sqrt{m_i + m_i^2(D - m_i)} \leq \frac{2D^3 + 9p + 2p^2\sqrt{D^2+3} + 6\sqrt{D^2+3}}{27},$$

where the final inequality is the closed-form solution of

$$\max_{m_i \in [0,D]} \sqrt{m_i + m_i^2(D - m_i)}.$$

□

H Proof of Lemma 5

Before the proof, we first restate Lemma 5 with a detained version.

Lemma 5 (Detailed Version). *Following the settings in Lemma 4, for any user $n \in [N]$, any $\rho \in (0,1)$, with probability at least $1 - \frac{(1+D+2e^2)\rho^2\pi^2}{6}$, for all sufficiently large $t \geq 8(2e^{2B}+1)\sqrt{2DT\log(\frac{T}{\rho})}$, the estimated preference of Eq.4 satisfies*

$$\|\hat{c}_t^{n(BF)} - \bar{c}^n\|_{V_t} \leq \alpha \frac{BD}{2} \sqrt{D \log\left(\frac{1+\alpha D(t-1)\ (\lambda+1)}{\delta}\right) + \log(\frac{1+\lambda}{\lambda\delta})} + \sqrt{1-\alpha}\left(\frac{D(8e^{4B}+4e^{2B})}{\sqrt{m_t^{\downarrow}}} \sqrt{\eta^2\left(\epsilon + 2\left(\frac{\epsilon\log(t/\rho)}{t}\right)^{1/2}\right) + \frac{4D\log(t/\rho)}{t}} + B\sqrt{2\lambda} \right).$$

Proof. The proof can be greatly simplified by applying the results in our previous proofs.

Specifically, by Eq. 13 in uncorrupted case, we have

$$\|\hat{c}_t^{(\mathrm{BF})} - \bar{c}\|_{V_t} \leq (1-\alpha) \underbrace{\left\| V_t^{-1} U_\lambda (\hat{c}_t^{(\mathrm{QE})} - \bar{c}) \right\|_{V_t}}_{A} + \alpha \underbrace{\left\| V_t^{-1} \sum_{\tau=1}^{t-1} r_{a_\tau,\tau} r_{a_\tau,\tau}^\top \xi_\tau \right\|_{V_t}}_{B},$$

where the term B is exactly same with the uncorrupted case since the independence with the corruption ϵ. For term A, by the derivation in Eq. 15, we have

$$A \leq \frac{1}{\sqrt{1-\alpha}} \left\| \hat{c}_t^{(\mathrm{QE})} - \bar{c} \right\|_{U_\lambda} \leq \frac{1}{\sqrt{1-\alpha}} \sqrt{(\hat{c}_t^{(\mathrm{QE})} - \bar{c}^0)^\top U (\hat{c}_t^{(\mathrm{QE})} - \bar{c}^0)} + B\sqrt{2\lambda D}.$$

Note the only difference is the new QE-estimator error bound induced by the ϵ-corruption model, as has been characterized in Lemma 5. Pluging the new QE-estimator error bound of Lemma 5 yields

$$\left\| \hat{c}_t^{(\mathrm{BF})} - \bar{c} \right\|_{V_t} \leq \alpha \frac{BD}{2} \sqrt{D \log\left(\frac{1+\alpha D(t-1)\ (\lambda+1)}{\delta}\right) + \log(\frac{1+\lambda}{\lambda\delta})}$$
$$+ \sqrt{1-\alpha}\left(\frac{D(8e^{4B}+4e^{2B})}{\sqrt{m_t^{\downarrow}}} \sqrt{\eta^2\left(\epsilon + 2\left(\frac{\epsilon\log(t/\rho)}{t}\right)^{1/2}\right) + \frac{4D\log(t/\rho)}{t}} + B\sqrt{2\lambda D} \right),$$

completing the proof.

□

I Proof of Theorem 4

Before the detailed proofs, we first restate Theorem 2 with a detained version and show how the optimal choices of balance-weight α and regularization λ are derived.

Theorem 4 (Detailed Version). *For Algorithm 1 with group-wise Lasso MLE, by setting β_t as the UCB of $\hat{\mathbf{c}}_t^{n(BF)}$ in Lemma 4, setting η as Lemma 4, then for any user $n \in [N]$, any sparse corruption rate $\epsilon \in [0,1]$, with probability at least $1 - \frac{(2+D+2e^2)\rho^2\pi^2}{6}$, we have*

$$R^n(T) \leq \Psi^{n(\hat{c})}(T) + \Psi^{n(\hat{r})}(T) + M, \quad \textit{where}$$

$$M = \max\left\{\frac{2D\alpha}{\nu_{r\downarrow}^2}\sqrt{KT\log(\tfrac{T}{\rho})}, \frac{4\epsilon\eta^4}{\alpha^2}\log(\tfrac{T}{\rho}), \frac{4D}{\alpha}\log(\tfrac{T}{\rho}), 8(2e^{2B}+1)\sqrt{2DT\log(\tfrac{T}{\rho})}\right\}$$

$$\Psi^{n(\hat{c})}(T) = O\Bigg(\Bigg(\frac{D^2\sqrt{D+\eta\epsilon/\alpha}}{\sqrt{m_T^{\downarrow}}} + BD\sqrt{\frac{\lambda}{\alpha}} + BD^{\frac{3}{2}}\sqrt{\alpha\log(\tfrac{T}{\rho}) + \alpha\log(\tfrac{1+\lambda}{\lambda\rho})}\Bigg)\sqrt{T\log(\tfrac{\alpha T}{\lambda})}\Bigg);$$

$$\Psi^{n(\hat{r})}(T) = O\Bigg(\sqrt{\frac{KD^4}{m_T^{\downarrow}\lambda}}\log^2 T + \Bigg(BD + \eta\sqrt{\frac{D^3\epsilon}{m_T^{\downarrow}\lambda}} + \frac{\alpha BD^{\frac{3}{2}}}{\sqrt{\lambda}}\sqrt{\log(\tfrac{T}{\rho}) + \log(\tfrac{1+\lambda}{\lambda\rho})}\Bigg)\sqrt{KT\log(\tfrac{T}{\rho})}\Bigg).$$

Remark 3. *The parameters α and λ can be optimally tuned. By setting $\alpha = \lambda = O(\epsilon + \log^{-1}(T/\rho))$, we have*

$$R^n(T) \leq O\Bigg(\Bigg(D^2\sqrt{\frac{D+\eta}{m_T^{\downarrow}}} + BD + \eta\sqrt{\frac{D^3K}{m_T^{\downarrow}}}\Bigg)\sqrt{T\log(T)} + B\sqrt{D^3K(\epsilon + \tfrac{1}{\log(T/\rho)})T}\cdot\log(\tfrac{T}{\rho})\Bigg)$$

Proof. The proof can be greatly simplified by applying the results in our previous proofs. Specifically, by Eq. 19 and Eq. 21, we have the regret as following formula.

$$\begin{aligned} R(T) &\leq O(M) + \underbrace{\sum_{t=M+1}^{T}\min\left(2B_{a_t,t}^{c}, BD\right)}_{R_{M+1:T}^{\hat{\mathbf{c}}}} + \underbrace{\sum_{t=M+1}^{T} 2\|\overline{\mathbf{c}}\|_1\sqrt{\frac{\log(t/\alpha)}{N_{a,t}}}}_{R_{M+1:T}^{\tilde{\mathbf{r}}}} \\ &\leq O(M) + 2\sqrt{2}\underbrace{\sum_{t=M+1}^{T}\min\left(\beta_t\|\boldsymbol{\mu}_{a_t}\|_{\mathbb{E}[\mathbf{V}_t]^{-1}}, \frac{BD}{2\sqrt{2}}\right)}_{\mathrm{Reg}^{(1)}} + 4\underbrace{\sum_{t=M+1}^{T}\beta_t\sqrt{\frac{D}{\lambda}}\gamma_{a_t,t}}_{\mathrm{Reg}^{(2)}} + R_{M+1:T}^{\tilde{\mathbf{r}}}. \end{aligned} \tag{38}$$

Note that the only difference is the new β_t induced by additional corruption component, leading to different $\mathrm{Reg}^{(1)}$ and $\mathrm{Reg}^{(2)}$. In the following, we show the derivation of these two terms while omit others since they are exactly the same with uncorrupted case.

Step-1.1. (Upper-Bound over $\mathrm{Reg}_t^{(1)}$)

Plugging the $\beta_t = \sqrt{1-\alpha}\Big(\frac{D(8e^{4B}+4e^{2B})}{\sqrt{m_t^{\downarrow}}}\sqrt{\eta^2\big(\epsilon + 2(\frac{\epsilon\log(t/\rho)}{t})^{1/2}\big) + \frac{4D\log(t/\rho)}{t}} + B\sqrt{2\lambda D}\Big) + \alpha\frac{BD}{2}\sqrt{D\log\big(\frac{1+\alpha D(t-1)\ (\lambda+1)}{\delta}\big) + \log(\frac{1+\lambda}{\lambda\delta})}$ (defined in Lemma 5), and by Cauchy–Schwarz inequality, we have

$$\begin{aligned} \mathrm{Reg}^{(1)} \leq\ & D(32e^{4B}+16e^{2B})\sqrt{\frac{D(1-\alpha)}{m_t^{\downarrow}}}\sqrt{\sum_{t=M+1}^{T}1\cdot\underbrace{\sum_{t=M+1}^{T}\min\left(\frac{\log(t/\rho)}{t}\|\boldsymbol{\mu}_{a_t}\|_{\mathbb{E}[\mathbf{V}_t]^{-1}}^2, \frac{B^2D^2}{8}\right)}_{I_1}} \\ & + D(32e^{4B}+16e^{2B})\sqrt{\frac{(1-\alpha)}{m_t^{\downarrow}}}\sqrt{\sum_{t=M+1}^{T}1\cdot\underbrace{\sum_{t=M+1}^{T}\min\left(2\eta^2\sqrt{\frac{\epsilon\log(t/\rho)}{t}}\|\boldsymbol{\mu}_{a_t}\|_{\mathbb{E}[\mathbf{V}_t]^{-1}}^2, \frac{B^2D^2}{8}\right)}_{I_*}} \\ & + \left(\eta D(32e^{4B}+16e^{2B})\sqrt{\frac{(1-\alpha)\epsilon}{m_T^{\downarrow}}} + B\sqrt{2\lambda(1-\alpha)} + \alpha\frac{BD}{2}\sqrt{D\log\left(\frac{1+\alpha DT/(\lambda+1)}{\rho}\right) + \log(\frac{1+\lambda}{\lambda\rho})}\right) \\ & \times\sqrt{\sum_{t=M+1}^{T}1\cdot\underbrace{\sum_{t=M+1}^{T}\min\left(\|\boldsymbol{\mu}_{a_t}\|_{\mathbb{E}[\mathbf{V}_t]^{-1}}^2, \frac{B^2D^2}{8}\right)}_{I_2}}. \end{aligned} \tag{39}$$

Note for I_1 and I_2, we have obtained the results in previous uncorrupted case. Next we show the bound of term I_*. When $\frac{4\epsilon\eta^4 \log \frac{t}{\rho}}{\alpha^2} \le t$, we have $2\eta^2\sqrt{\frac{\epsilon \log \frac{t}{\rho}}{t}} \le \alpha$, which gives

$$\begin{aligned} I_* &= \sum_{t=1}^{T} \min\left(2\eta^2\sqrt{\frac{\epsilon\log(t/\rho)}{t}}\|x\|^2_{V_t^{-1}}, \frac{B^2D^2}{8}\right) \\ &\le \sum_{t=1}^{T} \min\left(\alpha\|x\|^2_{V_t^{-1}}, \frac{B^2D^2}{8}\right) \\ &\le c_0 D\log\left(1+\frac{\alpha}{\lambda(1-\alpha)}(T-M)\right), \end{aligned}$$

where the last inequality holds by Eq. 23. Putting together, we have for $t \ge \max\{\frac{\log(T/\rho)}{\alpha}, \frac{4\epsilon\eta^4\log\frac{T}{\rho}}{\alpha^2} \le t\}$,

$$\begin{aligned} \mathrm{Reg}^{(1)} \le\ & D(32e^{4B}+16e^{2B})(D+\sqrt{D})\sqrt{\frac{c_0(1-\alpha)}{m_T^{\downarrow}}}\sqrt{(T-M)\log\left(1+\frac{\alpha}{\lambda(1-\alpha)}(T-M)\right)} \\ &+\eta D(32e^{4B}+16e^{2B})\sqrt{\frac{(1-\alpha)\epsilon}{\alpha m_T^{\downarrow}}}\sqrt{(T-M)\log\left(1+\frac{\alpha}{\lambda(1-\alpha)}(T-M)\right)} \\ &+\left(B\sqrt{\frac{2\lambda(1-\alpha)c_0D}{\alpha}}+\frac{BD}{2}\sqrt{\alpha c_0 D}\sqrt{D\log\left(\frac{1+\alpha DT/(\lambda+1)}{\rho}\right)+\log\left(\frac{1+\lambda}{\lambda\rho}\right)}\right) \\ &\times\sqrt{(T-M)\log\left(1+\frac{\alpha}{\lambda(1-\alpha)}(T-M)\right)}. \end{aligned} \tag{40}$$

Step-1.2. (Upper-Bound over $\mathrm{Reg}_t^{(2)}$)

Plugging the new β_t (defined in Lemma 5), we have

$$\begin{aligned} \mathrm{Reg}^{(2)} \le\ & D^2(32e^{4B}+16e^{2B})\sqrt{\frac{(1-\alpha)}{m_T^{\downarrow}\lambda}}\sum_{t=M+1}^{T}\sqrt{\frac{\log^2(t/\rho)}{tN_{a_t,t}}} \\ &+D(16e^{4B}+8e^{2B})\sqrt{\frac{D(1-\alpha)}{m_T^{\downarrow}\lambda}}\eta\sum_{t=M+1}^{T}\underbrace{\sqrt{\epsilon+2\left(\frac{\epsilon\log(t/\rho)}{t}\right)^{1/2}}}_{A}\sqrt{\frac{\log(t/\rho)}{N_{a_t,t}}} \\ &+\left(B\sqrt{2\lambda(1-\alpha)}+\alpha\frac{BD}{2}\sqrt{D\log\left(\frac{1+\alpha DT/(\lambda+1)}{\rho}\right)+\log\left(\frac{1+\lambda}{\lambda\rho}\right)}\right)\sqrt{\frac{D}{\lambda}}\sum_{t=M+1}^{T}\sqrt{\frac{\log(t/\rho)}{N_{a_t,t}}}. \end{aligned} \tag{41}$$

Note term A is the additional term introduced by corruption. We can see that $O(1) \le O(A) \le O(\frac{\log(t/\rho)}{t})$, thus we have the second term can be bounded as

$$O\left(D(16e^{4B}+8e^{2B})\sqrt{\frac{D\epsilon(1-\alpha)}{m_T^{\downarrow}\lambda}}\eta\sum_{t=M+1}^{T}\sqrt{\frac{\log(t/\rho)}{N_{a_t,t}}}\right) = O\left(\eta D\sqrt{\frac{D\epsilon}{\lambda}}\sqrt{KT\log\left(\frac{T}{\rho}\right)}\right).$$

Pluging above result back to Eq. 41, and combining Eq. 41, Eq. 41, Eq. 38 and Eq. 27 conclude the proof of $R(T)$. □

J Auxiliary Lemmas

Lemma 13 (Variant of Lemma 7 in [22]). *Assume that a bandit algorithm enjoys a sub-linear regret bound, then* $\mathbb{E}[N_{i,T}] = o(T), \forall i \ne a^*$.

Lemma 14 (Theorem 1.8 of [20]). *Suppose that* $S_r = \sum_{t=1}^{r} X_t$ *is a martingale where* $X_1, X_2, \ldots, X_m$ *take values in* $\mathbb{R}^n$ *and are such that* $\mathbb{E}[X_t] = 0$ *and* $\|X_t\|_2 \le D$ *for all* t*, for* $D > 0$*. Then, for every* $x > 0$*,*

$$\mathbb{P}\left[\|S_r\|_2 > x\right] \le 2e^2\exp\left(-\frac{x^2}{2rD^2}\right).$$

Lemma 15 (Hoeffding's inequality for general bounded random variables, Theorem 2.2.6 of [31]). *Given independent random variables* $\{X_1, ..., X_m\}$ *where* $a_i \le X_i \le b_i$ *almost surely (with probability 1) we have:*

$$\mathbb{P}\left(\frac{1}{m}\sum_{i=1}^{m}X_i - \frac{1}{m}\sum_{i=1}^{m}\mathbb{E}[X_i] \ge \epsilon\right) \le \exp\left(\frac{-2\epsilon^2m^2}{\sum_{i=1}^{m}(b_i-a_i)^2}\right).$$

Lemma 16 (Self-Normalized Bound for Vector-Valued Martingales [2]). *Let $\{\mathcal{F}_t\}_{t=0}^{\infty}$ be a filtration. Let $\{\eta_t\}_{t=1}^{\infty}$ be a real-valued stochastic process such that η_t is $\mathcal{F}_t$-measurable and conditionally R-sub-Gaussian for some $R \geq 0$, i.e.,*

$$\forall \lambda \in \mathbb{R}, \quad \mathbb{E}\left[e^{\lambda \eta_t} \mid \mathcal{F}_{t-1}\right] \leq \exp\left(\frac{\lambda^2 R^2}{2}\right).$$

Let $\{X_t\}_{t=1}^{\infty}$ be an $\mathbb{R}^d$-valued stochastic process such that X_t is $\mathcal{F}_{t-1}$-measurable. Assume that V is a $d \times d$ positive definite matrix. For any $t \geq 0$, define

$$\overline{V}_t = V + \sum_{s=1}^{t} X_s X_s^{\top}, \qquad S_t = \sum_{s=1}^{t} \eta_s X_s.$$

Then, for any $\delta > 0$, with probability at least $1-\delta$, for all $t \geq 0$,

$$\|S_t\|^2_{\overline{V}_t^{-1}} \leq 2R^2 \log\left(\frac{\det(\overline{V}_t)^{1/2}\det(V)^{-1/2}}{\delta}\right).$$

Lemma 17 (Determinant of Symmetric PSD Matrices Sum). *Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric and positive definite matrix, and $\boldsymbol{B} \in \mathbb{R}^{n\times n}$ be a symmetric and positive (semi-) definite matrix. Then we have*

$$\det(\boldsymbol{A}+\boldsymbol{B}) \geq \det(\boldsymbol{A}) + \det(\boldsymbol{B})$$

Proof.

$$\det(\boldsymbol{A}+\boldsymbol{B}) = \det(\boldsymbol{A})\det\left(\boldsymbol{I} + \boldsymbol{A}^{-\frac{1}{2}}\boldsymbol{B}\boldsymbol{A}^{-\frac{1}{2}}\right). \tag{42}$$

Let $\lambda_1, ..., \lambda_n$ be the eigenvalues of $\boldsymbol{A}^{-\frac{1}{2}}\boldsymbol{B}\boldsymbol{A}^{-\frac{1}{2}}$. Since $\boldsymbol{A}^{-\frac{1}{2}}\boldsymbol{B}\boldsymbol{A}^{-\frac{1}{2}}$ is positive (semi-) definite, we have $\lambda_i \geq 0, \forall i \in [n]$, which implies

$$\det\left(\boldsymbol{I} + \boldsymbol{A}^{-\frac{1}{2}}\boldsymbol{B}\boldsymbol{A}^{-\frac{1}{2}}\right) = \prod_{i=1}^{n}(1+\lambda_i) \geq 1 + \prod_{i=1}^{n}\lambda_i = \det(\boldsymbol{I}) + \det\left(\boldsymbol{A}^{-\frac{1}{2}}\boldsymbol{B}\boldsymbol{A}^{-\frac{1}{2}}\right). \tag{43}$$

Combining Eq.42 with Eq. 43 concludes the proof.

□

Lemma 18 (Lemma 10 in [9]). *Let $M = \left\lfloor \min\left\{t' \mid t v_{r\downarrow}^2 + \lambda \geq 2D\omega\sqrt{Kt\log\frac{t}{\alpha}}, \forall t \geq t'\right\}\right\rfloor$, under the conditions of Claim 1 and Claim 2, for any $t \geq M+1$, and any $\boldsymbol{\mu} \in \mathbb{R}^D$, we have*

$$\boldsymbol{\mu}^{\top}\boldsymbol{V}_{t-1}^{-1}\boldsymbol{\mu} \leq 2\boldsymbol{\mu}^{\top}\mathbb{E}[\boldsymbol{V}_{t-1}]^{-1}\boldsymbol{\mu}.$$

Lemma 19 (Lemma 12 in [9]). *For any $M \geq 0$, we have*

$$\sum_{t=M+1}^{T}\sqrt{\frac{\log\left(\frac{t}{\rho}\right)}{N_{a_t,t}}} \leq 2\sqrt{K(T-M)\log\left(\frac{T}{\rho}\right)}.$$

Lemma 20. *For any $M \geq 0$, we have*

$$\sum_{t=M+1}^{T}\sqrt{\frac{\log^2(t/\rho)}{tN_{a,t}}} = O\left(\sqrt{K}\cdot\log^2(T/\rho)\right).$$

Proof. By the Cauchy-Schwarz inequality, we have:

$$\sum_{t=M+1}^{T}\sqrt{\frac{\log^2(t/\rho)}{tN_{a_t,t}}} \leq \sqrt{\sum_{t=M+1}^{T}\frac{\log^2(t/\rho)}{t}}\cdot\sqrt{\sum_{t=M+1}^{T}\frac{1}{N_{a_t,t}}}.$$

We bound the first sum term by an integral:

$$\sum_{t=M+1}^{T}\frac{\log^2(t/\rho)}{t} \leq \int_M^T \frac{\log^2(x/\rho)}{x}\,dx.$$

Let $u = \log(x/\rho) \Rightarrow x = \rho e^u, dx = \rho e^u du$. Then:

$$\int_M^T \frac{\log^2(x/\rho)}{x}dx = \int_{\log(M/\rho)}^{\log(T/\rho)} u^2\,du = \left[\frac{u^3}{3}\right]_{\log(M/\rho)}^{\log(T/\rho)}.$$

So:

$$\sum_{t=M+1}^{T}\frac{\log^2(t/\rho)}{t} = O\left(\log^3(T/\rho)\right).$$

We then bound the second term. Note that a_t is the selected arm at time t, and for each time t, $N_{a_t,t}$ is incremented by 1. Thus:

$$\sum_{t=1}^{T} \frac{1}{N_{a_t,t}} \leq \sum_{i=1}^{K} \sum_{s=1}^{N_{i,T}} \frac{1}{s} \leq \sum_{i=1}^{K} \log N_{i,T} \leq K \log T,$$

which gives:

$$\sum_{t=M+1}^{T} \frac{1}{N_{a_t,t}} \leq O(K \log T).$$

Putting both bounds together:

$$\sum_{t=M+1}^{T} \sqrt{\frac{\log^2(t/\rho)}{t N_{a_t,t}}} \leq \sqrt{O(\log^3(T/\rho))} \cdot \sqrt{O(K \log T)} = O(\sqrt{K} \cdot \log^2(T/\rho)).$$

□

Lemma 21 (Standard Chernoff Bound (Additive Form))**.** *Let S_T be the sum of t i.i.d. Bernoulli random variables with mean ε. Then for any $\rho > 0$, the following inequality holds:*

$$\mathbb{P}\left(S_t \geq \varepsilon t + \sqrt{2\varepsilon t \log\left(\frac{1}{\rho}\right)}\right) \leq \rho.$$

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